Supplementary Material

For Paper "Near-Optimal Lower Bounds under ETH for Approximating k-means and k-median in ℓ_1 and ℓ_2 metrics" by Karthik C. S. and Dor Minzer

Organization of this Document

In Section 1, we prove the same conditional lower bounds against approximation schemes as in [KM26] for the Euclidean metric, but now in the ℓ_1 metric for k-means and k-median clustering objectives. In Section 2, we provide a corollary of the result in Section 1 against exact algorithms for k-means and k-median in the ℓ_1 metric. Finally, in Sections 3 and 4, we improve the results of [KM26] for Euclidean k-means and Euclidean k-median, respectively, to hold even when $k < 1/\varepsilon$.

Statement of AI Usage

Upon the arrival of ChatGPT 6, the authors asked it to extend the results in [KM26] to k-median and k-means in the ℓ_1 metric, as presented in Section 1. The model needed a hint to use a Hadamard matrix, as well as a suggestion to use small-bias sets to prove Theorem 2.1. Finally, for the results in Sections 3 and 4, the authors suggested that the properties of the explicit family of colorings in [KM26] could be relaxed and that the soundness analysis was robust enough to accommodate this with a modified construction. The model then worked out the details.

1 Conditional Lower Bounds for k-median and k-means in the ℓ_1 Metric under ETH

We now prove the corresponding conditional time lower bounds for k-median and k-means in the ℓ_1 metric in Sections 1.1 and 1.2 respectively. Similar to the previous section, our starting point is the hypergraph coverage problem in Theorem 2.2 in [KM26], and throughout this section, let L be the constant degree exponent supplied by that theorem with $\varepsilon = 1/100$.

1.1 Lower Bound for k-median in ℓ_1 Metric

We prove the following theorem in this subsection for k-median in ℓ_1-metric, and since the lower bound is for Boolean points, the result also applies to the Hamming metric (for categorical data).

Theorem 1.1. *Assuming ETH, there is a constant $C > 0$ such that the following holds. Let $\widetilde{\varepsilon}\colon \mathbb{N} \to (0,1)$, $\widetilde{k}\colon \mathbb{N} \to \mathbb{N}$ be polynomial time computable functions of the input size satisfying:*

1. $\frac{\widetilde{k}(n)}{\widetilde{\varepsilon}(n)} = o\left(\frac{n}{\log^L n}\right)$ *and the sequence* $\frac{\widetilde{k}(n)}{\widetilde{\varepsilon}(n)}$ *has bounded consecutive ratios.*
2. $\omega(\log n) \leqslant \frac{1}{\widetilde{\varepsilon}(n)} \leqslant o\left(\frac{1}{(\log n)^{2L}}\right)\widetilde{k}(n)$.

Then for sufficiently large n, setting $\varepsilon = \widetilde{\varepsilon}(n)$ and $k = \widetilde{k}(n)$, k-median in ℓ_1 metric on n points in $\{0,1\}^d$ has no $(1+\varepsilon)$-approximation running in time $2^{(k/\varepsilon)/(\log(k/\varepsilon))^C} \operatorname{poly}(n,d)$.

We begin with the Euclidean k-means instances P_ρ from Theorem 3.2, for which we already have the desired gap between the completeness and soundness cases. Our goal is to encode this gap into an instance of k-median in the ℓ_1 metric. However, we cannot simply view the same point sets under the ℓ_1 metric. Indeed, in both the completeness and soundness cases there is a simple clustering that ignores the vertex coordinates entirely. For every point p_e, choose any color $c(e) \in \rho(e)$ and group the points according to this choice. This gives at most k clusters, each with a common color c. Taking $\mathbf{e}_{N+c}$ as the center of such a cluster sets all N vertex coordinates to zero and gives ℓ_1 distance exactly $2t-1$ from every point in the cluster. Thus both cases admit a clustering of cost $(2t-1)|P_\rho|$ that is independent of the hypergraph structure responsible for the Euclidean k-means gap. Our goal is therefore to transform P_ρ into a new point set whose ℓ_1 median cost reflects the Euclidean k-means cost of P_ρ, so we can then use the conclusions of Theorem 3.2.

For simplicity, we first describe the construction in the one-dimensional setting, where the main ideas are easier to see. We start by the following fact.

Fact 1.2. *Let T be a sufficiently large constant and let U be a real-valued random variable uniformly distributed on $[-T,T]$. Then, for every b with $|b| \leqslant T$, we have $\mathbb{E}|U-b| = \frac{T}{2} + \frac{b^2}{2T}$.*

Proof.
$$\begin{aligned}
\mathbb{E}|U-b| &= \frac{1}{2T}\int_{-T}^{T} |u-b|\,du \\
&= \frac{1}{2T}\left(\int_{-T}^{b}(b-u)\,du + \int_{b}^{T}(u-b)\,du\right) \\
&= \frac{1}{2T}\left(\frac{(T+b)^2}{2} + \frac{(T-b)^2}{2}\right) \\
&= \frac{T^2+b^2}{2T} = \frac{T}{2} + \frac{b^2}{2T}.
\end{aligned}$$
□

Our use of uniform random shifts is inspired by the randomized-smoothing construction of Duchi, Bartlett, and Wainwright [DBW12], where averaging the absolute-value loss under a uniform perturbation yields a quadratic function in the relevant range.

To see why this fact is useful, consider a point $z \in \mathbb{R}$ and a center $\mu \in \mathbb{R}$, and shift the point z by an independent copy of U. Its distance to μ becomes $|(z+U)-\mu| = |U-(\mu-z)|$. Therefore, provided $|\mu - z| \leqslant T$, Fact 1.2 gives $\mathbb{E}\left[|(z+U)-\mu|\right] = \frac{T}{2} + \frac{(z-\mu)^2}{2T}$. Thus, after adding the random shift and taking expectation, the ℓ_1 distance from $z+U$ to μ is equal to a common additive term $T/2$ plus a scaled version of the squared distance from the original point z to μ. Thus, adding a uniform random shift converts an absolute distance into a quadratic function of the displacement.

More generally, suppose $z_1,\dots,z_m \in [-R,R]$ for some $R > 0$, and let $\mu = m^{-1}\sum_{j\in[m]} z_j$ be their mean. Since $\mu \in [-R,R]$, we have $|\mu - z_j| \leqslant 2R$ for every $j \in [m]$. Thus, provided $T \geqslant 2R$, if $U_1,\dots,U_m$ are independent copies of U and we shift each z_j to $z_j + U_j$, then Fact 1.2 gives,

$$\mathbb{E}\left[\sum_{j\in[m]} |z_j + U_j - \mu|\right] = \frac{mT}{2} + \frac{1}{2T}\sum_{j\in[m]} (z_j-\mu)^2. \tag{1}$$

Note that the first term depends only on the number of points, while the second term is precisely the one-dimensional 1-means cost, scaled by $1/(2T)$. This suggests that, after applying the trans-

formation in sufficiently many one-dimensional directions, the resulting ℓ_1 cost should recover the Euclidean k-means cost.

At this point, however, the above observation does not yet give the desired construction. There are two issues that we need to address.

First, (1) computes the expected cost of the shifted points when we use the particular center μ. This is not necessarily their optimal ℓ_1 cost. Indeed, once the shifts $U_1, \dots, U_m$ are fixed, the optimal center is a median of the shifted points $z_1 + U_1, \dots, z_m + U_m$. Thus, what we actually need to understand is the optimal ℓ_1 cost of the shifted points. For a fixed choice of the shifts $U_1, \dots, U_m$, the optimal ℓ_1 cost of the shifted points is $\min_{b \in \mathbb{R}} \sum_{j \in [m]} |z_j + U_j - b|$. Taking expectation over the random shifts, the expected optimal cost is therefore $\mathbb{E}\left[\min_{b \in \mathbb{R}} \sum_{j \in [m]} |z_j + U_j - b|\right]$. We will show that the saving obtained by choosing the optimal center after seeing the shifts, rather than fixing the center to be μ, is small. More precisely,

$$0 \leqslant \mathbb{E}\left[\sum_{j \in [m]} |z_j + U_j - \mu|\right] - \mathbb{E}\left[\min_{b \in \mathbb{R}} \sum_{j \in [m]} |z_j + U_j - b|\right] \leqslant 6T.$$

Importantly, this saving is bounded by $6T$ regardless of the number m of points in the cluster. Hence, over a clustering into at most k clusters, the total saving is only $O(Tk)$.

Second, the preceding discussion only applies to one-dimensional points $z_1, \dots, z_m$, whereas the points $p_1, \dots, p_m$ in P_ρ lie in $\mathbb{R}^{d_0}$ (where $d_0 = \mathrm{poly}(m)$). To reduce the high-dimensional problem to the one-dimensional setting, we use the Walsh–Hadamard transform. More precisely, we obtain a collection $\mathcal{G}$ of sign vectors $g \in \{-1, 1\}^{d_0}$ with the property that, for every $v \in \mathbb{R}^{d_0}$,

$$\frac{1}{|\mathcal{G}|} \sum_{g \in \mathcal{G}} \langle g, v \rangle^2 = \|v\|_2^2.$$

This is simply the orthogonality property of the rows of a Hadamard matrix: when the squares are expanded and averaged, all cross terms cancel.

For each $g \in \mathcal{G}$, we now project the original points in P_ρ onto the one-dimensional direction g by setting $z_j(g) := \langle g, p_j \rangle$. Let $\overline{p} = m^{-1} \sum_{j \in [m]} p_j$ be the centroid of the original points. The mean of the projected points $z_1(g), \dots, z_m(g)$ is then $\langle g, \overline{p} \rangle$, and hence their one-dimensional 1-means cost is $\sum_{j \in [m]} \big(z_j(g) - \langle g, \overline{p} \rangle\big)^2 = \sum_{j \in [m]} \langle g, p_j - \overline{p} \rangle^2$. Averaging over all Hadamard sign vectors and using the orthogonality property above gives us:

$$\frac{1}{|\mathcal{G}|} \sum_{g \in \mathcal{G}} \sum_{j \in [m]} \big(z_j(g) - \langle g, \overline{p} \rangle\big)^2 = \sum_{j \in [m]} \frac{1}{|\mathcal{G}|} \sum_{g \in \mathcal{G}} \langle g, p_j - \overline{p} \rangle^2 = \sum_{j \in [m]} \|p_j - \overline{p}\|_2^2.$$

The last expression is exactly the Euclidean 1-means cost of the original cluster. Thus, the Hadamard transform allows us to let the Euclidean squared cost as the average of the one-dimensional squared costs considered above. We can therefore apply the random-shift construction separately to each Hadamard projection.

It remains to make the random shifts finite and deterministic, which we do constructing a finite family of shifts having the limited independence needed for the analysis. The coordinates of the new point set will be indexed by a choice of Hadamard sign vector together with a choice from this finite family of shifts. Averaging over these coordinates then plays the role of averaging

over both the Hadamard projections and the random shifts. As a result, the ℓ_1 k-median cost of the new point set recovers, up to a common additive term and a term of magnitude $O(T)$ for each cluster, a scaled version of the Euclidean k-means cost of P_ρ.

1.1.1 Construction

Fix a retained coloring ρ, and list the points of P_ρ as $p_1, \ldots, p_m$, counting repeated hyperedges separately. Recall that the point corresponding to a hyperedge e has a one in each of the t vertex coordinates of e and in each of its t color coordinates, and has zeros elsewhere. Thus $p_j \in \{0,1\}^{N+k}$ and $\|p_j\|_1 = 2t$ for every $j \in [m]$. We describe the transformation for a general multiset of integer points $P = \{p_1, \ldots, p_m\} \subseteq \mathbb{Z}^{d_0}$ satisfying $\|p_j\|_1 \leqslant R$, where R is a positive integer. For the hypergraph points, we will take $d_0 = N + k$ and $R = 2t$.

We first specify the Hadamard sign vectors that will be used to map each point $p_j \in \mathbb{R}^{d_0}$ to a collection of one-dimensional values. Let $h \geqslant d_0$ be the smallest power of two. The Hadamard matrix of order h is obtained from the recursion $H_1 = (1)$ and $H_{2u} = \begin{pmatrix} H_u & H_u \\ H_u & -H_u \end{pmatrix}$, for every power of two u. All its entries are in $\{-1, 1\}$, and the recursion gives $H_h^{\mathsf{T}} H_h = h I_h$. Let $\mathcal{G}$ be the collection obtained by restricting each of its h rows to the first d_0 coordinates. Equivalently, we can append zeros to the original points to let them in $\mathbb{R}^h$, and then project onto the full rows of H_h. Appending zeros does not change any of their Euclidean distances. The orthogonality of the columns gives $h^{-1} \sum_{g \in \mathcal{G}} g(\ell) g(\ell') = 0$ for distinct $\ell, \ell' \in [d_0]$, while it evaluate to one when $\ell = \ell'$. Consequently, for every $v \in \mathbb{R}^{d_0}$,

$$\frac{1}{h} \sum_{g \in \mathcal{G}} \langle g, v \rangle^2 = \sum_{\ell, \ell' \in [d_0]} v_\ell v_{\ell'} \frac{1}{h} \sum_{g \in \mathcal{G}} g(\ell) g(\ell') = \sum_{\ell \in [d_0]} v_\ell^2 = \|v\|_2^2. \tag{2}$$

For every $g \in \mathcal{G}$, define the projected point $z_j(g) := \langle g, p_j \rangle$. Since the entries of g have absolute value one, we have $|z_j(g)| \leqslant \|p_j\|_1 \leqslant R$. Thus all these one-dimensional points lie in the interval required for the random-shift argument above.

We next specify a finite family of shifts. Fix an integer $T \geqslant 4R$, and let $q \geqslant m$ be a power of two. Divide $[-T, T]$ into q equal subintervals and use their midpoints as the possible shifts, i.e., for each $u \in \{0, \ldots, q-1\}$, we consider the shift $-T + \frac{(2u+1)T}{q}$. Every value in a subinterval is within T/q of its midpoint. Therefore we can compare these finite shifts with continuous uniform shifts by moving each point by at most T/q and this will be accounted for the resulting change in cost in the analysis.

If we chose the shift of each of the m points independently from the q possible values, there would be q^m possible choices, and using one coordinate for every choice would make the dimension prohibitively large. Fortunately, full independence is unnecessary. As will become clear in the analysis, the bound on the saving obtained by choosing a median uses only moments involving at most four shifts. It therefore suffices to use a *four-wise independent* family of shifts. Since $q \geqslant m$ is a power of two, the standard finite-field construction of four-wise independent functions gives a family $\mathcal{A}$ of q^4 functions $a : [m] \to [q]$ such that, when a is chosen uniformly from $\mathcal{A}$, the values $a(j_1), \ldots, a(j_r)$ are independent and uniform in $[q]$ for every $r \leqslant 4$ and every choice of distinct $j_1, \ldots, j_r \in [m]$; see, e.g., [Vad12].

For every $a \in \mathcal{A}$ and $j \in [m]$, define $\xi_j(a) := -T + \frac{T}{q}\big(2a(j) - 1\big)$. Thus $\xi_j(a)$ is one of the q

midpoints of the equal subintervals of $[-T, T]$. When a is chosen uniformly from $\mathcal{A}$, each $\xi_j(a)$ is uniform over these q midpoints, and the shifts associated with any four distinct point occurrences are independent.

We can now define the new points. There is one coordinate for every pair $(g, a) \in \mathcal{G} \times \mathcal{A}$, and the point y_j corresponding to p_j has coordinate:

$$(y_j)_{g,a} := q \cdot \big(z_j(g) + \xi_j(a)\big) = q\langle g, p_j\rangle + T\big(2a(j) - 1 - q\big). \tag{3}$$

The factor q removes the denominator in the shift, so all coordinates are integers. In particular, if p_j corresponds to the hyperedge e_j, then the point y_j at coordinate (g, a) can be written as:

$$(y_j)_{g,a} = q\left(\sum_{v \in e_j} g(v) + \sum_{c \in \rho(e_j)} g(N + c)\right) + T\big(2a(j) - 1 - q\big).$$

Let $Y_\rho := \{y_1, \ldots, y_m\}$. It has the same number of points as P_ρ, and dimension $D := hq^4 = O(d_0 q^4)$. Every coordinate has absolute value at most $q(R + T)$. For our reduction, we choose $T = 8t$ and take q to be the least power of two at least M, where M is the number of hyperedges before the coloring discards any of them. Thus T is a constant, $q = O(M)$, and the points can be written down in polynomial time.

Finally, we convert the integer points to Boolean points using the standard unary encoding of nonnegative integers, which gives an isometric embedding of ℓ_1 into Hamming space. Namely, after translating each coordinate ℓ so that its values lie in $\{0, \ldots, B_\ell\}$ for some $B_\ell \in \mathbb{N}$, we replace a value a by a block consisting of a ones followed by $B_\ell - a$ zeros. Two values a and b then differ in exactly $|a - b|$ positions in this block, and hence the encoding preserves the ℓ_1 distance between every pair of integer points. Moreover, an optimal ℓ_1 median of integer points can be chosen coordinate-wise to have integer coordinates. The encoding therefore preserves the optimal ℓ_1 median cost of every cluster, and hence the k-median optimum.

1.1.2 Analysis

We first address the saving obtained by choosing the optimal ℓ_1 median after seeing the shifts. As discussed above, (1) gives the expected ℓ_1 1-median cost of the shifted one-dimensional points when we use the particular center μ, where the resulting expression is written in terms of the one-dimensional 1-means cost of the original points. The following lemma shows that allowing the center to move from μ to the optimal median can decrease this expected ℓ_1 cost by at most $6T$, even when the shifts are only four-wise independent.

Lemma 1.3. *Let $R > 0$, $T \geqslant 4R$, and $z_1, \ldots, z_m \in [-R, R]$, where $m \geqslant 1$. Let $U_1, \ldots, U_m$ be four-wise independent random variables, each uniformly distributed on $[-T, T]$, and let $\mu = m^{-1}\sum_{j\in[m]} z_j$. Then,*

$$\frac{mT}{2} + \frac{1}{2T}\sum_{j\in[m]}(z_j - \mu)^2 - 6T \leqslant \mathbb{E}\left[\min_{b\in\mathbb{R}} \sum_{j\in[m]} |z_j + U_j - b|\right] \leqslant \frac{mT}{2} + \frac{1}{2T}\sum_{j\in[m]}(z_j - \mu)^2.$$

We apply this lemma separately for each fixed Hadamard direction $g \in \mathcal{G}$. For such a g, the values $z_j(g) = \langle g, p_j\rangle$ form a one-dimensional point set, while the coordinates (g, a) with $a \in \mathcal{A}$

correspond to the different choices of shifts for these projected points. Averaging over $a \in \mathcal{A}$ therefore plays the role of taking expectation over the shifts in Lemma 1.3.

Our construction uses the midpoint shifts $\xi_j(a)$ rather than continuous uniform shifts on $[-T, T]$. However, moving a point within its subinterval changes its position by at most T/q. By the triangle inequality, replacing all the shifts in a cluster $S \subseteq [m]$ by continuous uniform shifts changes its optimal ℓ_1 median cost by at most $T|S|/q$.

We also need to account for the scaling in (3). Fix a nonempty cluster $S \subseteq [m]$ and let $P_S := \{p_j : j \in S\}$ and $Y_S := \{y_j : j \in S\}$. An ℓ_1 center can be optimized independently in each coordinate. In coordinate (g, a), dividing both the points and the center by q gives the one-dimensional median problem on $\{z_j(g) + \xi_j(a) : j \in S\}$. Consequently,

$$\frac{\mathrm{OPT}_1^{(1,1)}(Y_S)}{qD} = \frac{1}{hq^4} \sum_{g \in \mathcal{G}} \sum_{a \in \mathcal{A}} \min_{b \in \mathbb{R}} \sum_{j \in S} |z_j(g) + \xi_j(a) - b|. \tag{4}$$

Thus dividing by qD accounts for both the factor q in each coordinate and the sum over all $D = hq^4$ coordinates. The right side is the average median cost over the Hadamard projections and the finite family of shifts. The preceding lemma compares each projected median cost with the common term $|S|T/2$ plus a $1/(2T)$ multiple of the projected 1-means cost. Averaging the latter cost over $g \in \mathcal{G}$ gives the Euclidean 1-means cost, by (2). Accounting also for the midpoint shifts and summing over clusters gives the following lemma.

Lemma 1.4. *Let $P = \{p_1, \dots, p_m\} \subseteq \mathbb{Z}^{d_0}$ satisfy $\|p_j\|_1 \leqslant R$ for every $j \in [m]$, and let Y be the point set obtained from (3) with $T \geqslant 4R$ and $q \geqslant m$. Let $D = hq^4$ denote the dimension of Y. Then, for every $1 \leqslant k \leqslant m$,*

$$\left| \frac{\mathrm{OPT}_k^{(1,1)}(Y)}{qD} - \left(\frac{mT}{2} + \frac{\mathrm{OPT}_k^{(2,2)}(P)}{2T} \right) \right| \leqslant 7Tk.$$

Before proving the above two lemmas, we use them to prove Theorem 1.1.

Proof of Theorem 1.1. Suppose that the asserted algorithm exists for a sufficiently large constant C. Given the hypergraph $\mathcal{H}$ from Theorem 2.2 in [KM26], choose n, k, ε, and $k' = k - 1$ as in the proof of Theorem 3.1, with the constant B chosen below. Let Δ denote the degree of $\mathcal{H}$ and let $M = N\Delta/t$. The earlier proof o Theorem 3.1 verifies the assumptions of Theorem 3.2 with k' clusters and gives $k/\varepsilon = \Theta(N)$, $k' \geqslant B\varepsilon N/2$, and $M = o(n)$.

For every retained coloring ρ, apply Lemma 1.4 to P_ρ with $R = 2t$, $d_0 = N + k'$, and $T = 8t$. Let q be the least power of two that is at least M, so that the same q and dimension $D = hq^4$ are used for all colorings. We obtain:

$$\left| \frac{\mathrm{OPT}_{k'}^{(1,1)}(Y_\rho)}{qD} - \left(\frac{|P_\rho|T}{2} + \frac{\mathrm{OPT}_{k'}^{(2,2)}(P_\rho)}{2T} \right) \right| \leqslant 7Tk'.$$

We now show that the bound $7Tk'$ is small compared with the completeness and soundness gap, and that the number of retained points does not affect this gap.

Let $m = |P_\rho| = M - B_\rho$, where $B_\rho = O_t(M/k')$ is the number of discarded hyperedge occur-

rences. Since $k' \geqslant 2t\Delta\sqrt{N}$, we have:

$$\frac{B_\rho}{k'\Delta} = O_t\left(\frac{N}{(k')^2}\right) = O_t(\Delta^{-2}) = o(1).$$

Thus $B_\rho = o(k'\Delta)$. Also $7Tk' = o(k'\Delta)$, because T is constant and $\Delta = \omega(1)$. To express the cost relative to a common value for all colorings, let $b_t := T/2 + (2t-1)/(2T)$. Then,

$$\frac{mT}{2} + \frac{\mathrm{OPT}_{k'}^{(2,2)}(P_\rho)}{2T} = b_t M - \frac{TB_\rho}{2} - \frac{(2t-1)M - \mathrm{OPT}_{k'}^{(2,2)}(P_\rho)}{2T}.$$

The last numerator is exactly the Euclidean cost saving bounded in Theorem 3.2. In the completeness case, some coloring has saving at least $(1-o(1))k'\Delta$. In the soundness case, every coloring has saving at most $(1-1/(8t))k'\Delta$. The term involving B_ρ is nonpositive and can be omitted for the completeness case upper bound; for the soundness case lower bound, it contributes only $o(k'\Delta)$. Therefore we have,

$$\frac{\mathrm{OPT}_{k'}^{(1,1)}(Y_\rho)}{qD} \leqslant b_t M - \frac{k'\Delta}{2T} + o(k'\Delta) \qquad \text{for some } \rho \text{ in completeness case,}$$

$$\frac{\mathrm{OPT}_{k'}^{(1,1)}(Y_\rho)}{qD} \geqslant b_t M - \left(1 - \frac{1}{8t}\right)\frac{k'\Delta}{2T} - o(k'\Delta) \qquad \text{for every } \rho \text{ in soundness case.}$$

Since t and T are fixed, take N large enough that each $o(k'\Delta)$ term has magnitude at most $k'\Delta/(128tT)$, and define $U := b_t M - (1-1/(64t))k'\Delta/(2T)$. The completeness upper bound is then at most U. The soundness lower bound exceeds U by at least

$$\left(\frac{1}{8t} - \frac{1}{64t}\right)\frac{k'\Delta}{2T} - \frac{k'\Delta}{128tT} = \frac{3k'\Delta}{64tT} > \frac{k'\Delta}{32tT}.$$

To distinguish the cases with a $(1+\varepsilon)$-approximation, we need this gap to exceed εU. Since $U \leqslant b_t M = b_t N\Delta/t$ and $k' \geqslant B\varepsilon N/2$, we have $\varepsilon U \leqslant b_t \varepsilon N\Delta/t$ and $k'\Delta/(32tT) \geqslant B\varepsilon N\Delta/(64tT)$. Choosing $B > 64Tb_t = 32(T^2+2t-1)$ therefore gives $(1+\varepsilon)U < U + k'\Delta/(32tT)$.

We now duplicate and pad the instance to obtain exactly n points and k centers, exactly as in the proof of Theorem 3.1. The only difference is that here we use the ℓ_1 triangle inequality to separate the padding point from the original points. More precisely, denoting $W := \lfloor (n-1)/m \rfloor$ and $A_0 := (T+2t)qD$, we take W copies of every point of Y_ρ and add $n - Wm$ copies of $x := 4nA_0\mathbf{e}_{\text{new}}$ in a fresh coordinate. Since every original point has ℓ_1 norm at most A_0, the same argument gives $\mathrm{OPT}_k^{(1,1)}(Y'_\rho) = W \cdot \mathrm{OPT}_{k-1}^{(1,1)}(Y_\rho)$. Finally, applying the unary encoding described above gives exactly n Boolean points with the same optimum and polynomial dimension.

Run the asserted algorithm on every Boolean instance, and divide each returned cost by WqD. In the completeness case, at least one coloring gives a cost at most $(1+\varepsilon)U$. In the soundness case, every returned cost is greater than this threshold. The number of colorings and the polynomial time required to construct the instances give the same total running-time bound as in the proof of Theorem 3.1, using $k/\varepsilon = \Theta(N)$. We thereby distinguish the two hypergraph cases within the time forbidden by Theorem 2.2 in [KM26], a contradiction. □

We will now prove Lemmas 1.3 and 1.4.

Proof of Lemma 1.3. Let $\widetilde{z}_j := z_j + U_j$ and, for every $b \in \mathbb{R}$, define $F(b) := \sum_{j\in[m]} |\widetilde{z}_j - b|$ to be the 1-median cost of the shifted points at center b. Since $|z_j - \mu| \leqslant 2R \leqslant T$ for every $j \in [m]$, Fact 1.2 gives us:

$$\mathbb{E}F(\mu) = \mathbb{E}\left[\sum_{j\in[m]} |z_j + U_j - \mu|\right] = \frac{mT}{2} + \frac{1}{2T}\sum_{j\in[m]} (z_j - \mu)^2. \tag{5}$$

The upper bound of the lemma follows immediately, since for every choice of the shifts the optimal 1-median cost is at most $F(\mu)$. For the lower bound, we show that moving from μ to a median saves at most $6T$ in expectation.

For every $b \in \mathbb{R}$, let $G(b) := \sum_{j\in[m]} (2\mathbf{1}_{\{\widetilde{z}_j \leqslant b\}} - 1)$, which is the number of shifted points at or to the left of b minus the number to its right. This is the derivative of the 1-median cost $F(b)$ with respect to the center location b: a shifted point to the left of b contributes 1, while a shifted point to the right contributes -1. The function G is nondecreasing. If we move the center to the right from μ, the cost decreases while G is negative and stops decreasing once G is nonnegative. Thus the saving in this direction is the integral of the negative part of G. Moving to the left gives the corresponding integral of its positive part. Since G is nondecreasing, at most one of these directions gives a positive saving. Let $x_+ := \max\{x, 0\}$, we therefore have:

$$F(\mu) - \min_{b\in\mathbb{R}} F(b) = \int_0^\infty (-G(\mu + s))_+ \, ds + \int_0^\infty (G(\mu - s))_+ \, ds. \tag{6}$$

We will bound the expectation of each integral by $3T$.

We first bound the variance and fourth moment of $G(b)$ for a fixed b. Subtract the expectation from each summand of $G(b)$. These centered summands are four-wise independent, have mean zero and variance at most 1, and have absolute value at most 2. Their fourth moments are consequently at most 4 times their variances, and hence at most 4. When we expand the fourth power of their sum, the only nonzero terms are the individual fourth powers and products of two squared summands. Therefore, $\mathrm{Var}(G(b)) \leqslant m$ and $\mathbb{E}\left[(G(b) - \mathbb{E}G(b))^4\right] \leqslant 4m + 6\binom{m}{2} \leqslant 4m^2$. This is the only estimate that requires four-wise independence.

We now bound the first integral in (6), which measures the possible saving from moving the center to the right of μ. For each $s > 0$, its integrand is $(-G(\mu + s))_+$. Thus we need to bound the probability and magnitude of the event that the derivative $G(\mu + s)$ is negative.

We first consider $0 < s \leqslant T/2$. In this range, the expectation of $G(\mu + s)$ can be computed exactly. Since $|\mu - z_j| \leqslant 2R \leqslant T/2$, we have $\mu + s - z_j \in [-T, T]$ for every $j \in [m]$. Therefore, using the uniform distribution of U_j and $\sum_{j\in[m]} (\mu - z_j) = 0$, we obtain,

$$\mathbb{E}G(\mu + s) = \sum_{j\in[m]} \left(2\Pr[U_j \leqslant \mu + s - z_j] - 1\right) = \frac{1}{T}\sum_{j\in[m]} (\mu + s - z_j) = \frac{ms}{T}.$$

Thus the derivative has positive expectation ms/T. For the cost to decrease at $\mu + s$, we must nevertheless have $G(\mu + s) < 0$, which requires $G(\mu + s)$ to deviate below its expectation by at least ms/T.

Let $X := G(\mu + s) - \mathbb{E}G(\mu + s)$. We use two bounds for the expected negative part. First, $(-G(\mu + s))_+ \leqslant |X|$, and Cauchy–Schwarz gives $\mathbb{E}[(-G(\mu + s))_+] \leqslant \mathbb{E}|X| \leqslant \sqrt{\mathbb{E}X^2} \leqslant \sqrt{m}$. Second, if $G(\mu + s) < 0$, then $|X| \geqslant ms/T$. On this event, $|X| \leqslant |X|^4/(ms/T)^3$, so the fourth-

moment bound gives:

$$\mathbb{E}[(-G(\mu+s))_+] \leqslant \mathbb{E}\left[|X|\mathbf{1}_{\{|X|\geqslant ms/T\}}\right] \leqslant \frac{\mathbb{E}X^4}{(ms/T)^3} \leqslant \frac{4T^3}{ms^3}.$$

Combining the bounds, we have $\mathbb{E}[(-G(\mu+s))_+] \leqslant \min\left\{\sqrt{m}, \frac{4T^3}{ms^3}\right\}$. The first bound controls the integral near $s = 0$, and the second decays as s increases. They agree at $s_0 := 4^{1/3}T/\sqrt{m}$. Extending the integration interval to all positive s gives:

$$\begin{aligned}\int_0^{T/2} \mathbb{E}[(-G(\mu+s))_+]\,ds &\leqslant \int_0^{s_0} \sqrt{m}\,ds + \int_{s_0}^{\infty} \frac{4T^3}{ms^3}\,ds \\ &= \sqrt{m}\,s_0 + \frac{2T^3}{ms_0^2} = \frac{3}{2}4^{1/3}T < \frac{5T}{2}.\end{aligned}$$

For $s \geqslant T/2$, monotonicity gives $\mathbb{E}G(\mu+s) \geqslant \mathbb{E}G(\mu+T/2) = m/2$. We now use the variance bound. For every real x and every $a > 0$, we have $(-x)_+ \leqslant (x-a)^2/(4a)$: if $x < 0$, this follows from $(x+a)^2 \geqslant 0$, and otherwise the left side is zero. Apply this with $x = G(\mu+s)$ and $a = \mathbb{E}G(\mu+s)$ and we obtain:

$$\mathbb{E}[(-G(\mu+s))_+] \leqslant \frac{\mathrm{Var}(G(\mu+s))}{4\mathbb{E}G(\mu+s)} \leqslant \frac{m}{4(m/2)} = \frac{1}{2}.$$

Every shifted point is at most $T+R$, while $\mu \geqslant -R$. Hence $G(\mu+s) = m$ once $s \geqslant T+2R \leqslant 3T/2$, and its negative part then vanishes. It follows that

$$\int_{T/2}^{\infty} \mathbb{E}[(-G(\mu+s))_+]\,ds \leqslant \int_{T/2}^{3T/2} \frac{1}{2}\,ds = \frac{T}{2}.$$

The expected saving from movement to the right is therefore at most $3T$. Reflecting all points about zero gives the same bound for movement to the left. Substituting both bounds into (6), and then using (5), proves the lower bound. □

Proof of Lemma 1.4. Fix a nonempty cluster with indices $S \subseteq [m]$. Let $P_S := \{p_j : j \in S\}$ and $Y_S := \{y_j : j \in S\}$, and let $\overline{p}_S := |S|^{-1}\sum_{j\in S} p_j$ be the centroid of P_S. We compare $\mathrm{OPT}_1^{(1,1)}(Y_S)/(qD)$ with $|S|T/2 + \mathrm{OPT}_1^{(2,2)}(P_S)/(2T)$.

By (4), the first quantity is the average projected median cost over all $(g,a) \in \mathcal{G}\times\mathcal{A}$. Fix $g \in \mathcal{G}$, and choose a uniformly from $\mathcal{A}$. To compare the midpoint shifts with continuous uniform shifts, independently choose ζ_j uniformly from $[-T/q, T/q]$ for every $j \in S$, and set $U_j := \xi_j(a) + \zeta_j$. The midpoint selects a uniform subinterval of $[-T,T]$, and ζ_j selects a uniform position within it. Thus each U_j is uniform on $[-T,T]$. The variables U_j are four-wise independent, because any four distinct midpoints are independent and the variables ζ_j are independent of each other and of a.

For every choice of these shifts and every center $b \in \mathbb{R}$, the triangle inequality gives:

$$\left|\sum_{j\in S}|z_j(g)+\xi_j(a)-b| - \sum_{j\in S}|z_j(g)+U_j-b|\right| \leqslant \sum_{j\in S}|\zeta_j| \leqslant \frac{T|S|}{q}.$$

This bound holds for every b, so it also holds for the two minimum costs. Taking expectations, we

obtain:

$$\left|\mathbb{E}_a\left[\min_{b\in\mathbb{R}}\sum_{j\in S}|z_j(g)+\xi_j(a)-b|\right]-\mathbb{E}\left[\min_{b\in\mathbb{R}}\sum_{j\in S}|z_j(g)+U_j-b|\right]\right|\leqslant\frac{T|S|}{q}. \tag{7}$$

All projected points lie in $[-R,R]$, so Lemma 1.3 applies to the second expectation. Their mean is $\langle g,\overline{p}_S\rangle$, and their squared deviations sum to $\sum_{j\in S}\langle g,p_j-\overline{p}_S\rangle^2$. Averaging this sum over $g\in\mathcal{G}$ and using (2) gives:

$$\frac{1}{h}\sum_{g\in\mathcal{G}}\sum_{j\in S}\langle g,p_j-\overline{p}_S\rangle^2=\sum_{j\in S}\|p_j-\overline{p}_S\|_2^2=\mathrm{OPT}_1^{(2,2)}(P_S).$$

The final equality holds because the centroid is an optimal Euclidean 1-means center. Combining Lemma 1.3, (7), and this identity, we obtain:

$$\begin{aligned}\frac{|S|T}{2}+\frac{\mathrm{OPT}_1^{(2,2)}(P_S)}{2T}-6T-\frac{T|S|}{q}&\leqslant\frac{\mathrm{OPT}_1^{(1,1)}(Y_S)}{qD}\\&\leqslant\frac{|S|T}{2}+\frac{\mathrm{OPT}_1^{(2,2)}(P_S)}{2T}+\frac{T|S|}{q}.\end{aligned}$$

Here $6T$ accounts for the saving from choosing a median, and $T|S|/q$ accounts for replacing continuous shifts by midpoints.

Summing the preceding bounds over k clusters, the $6T$ term per cluster contributes at most $6Tk$ in total, while the terms $T|S|/q$ sum to $Tm/q\leqslant T$. Their combined contribution is therefore at most $7Tk$. Since the additive term $mT/2$ is independent of the partition, minimizing over all partitions gives the claimed bound. □

1.2 Lower Bound for k-means in ℓ_1 Metric

We prove the following theorem in this subsection for k-means in ℓ_1-metric, and since the lower bound is for Boolean points, the result also applies to the Hamming metric.

Theorem 1.5. *Assuming ETH, there is a constant $C>0$ such that the following holds. Let $\widetilde{\varepsilon}\colon\mathbb{N}\to(0,1)$, $\widetilde{k}\colon\mathbb{N}\to\mathbb{N}$ be polynomial time computable functions of the input size satisfying:*

1. *$\frac{\widetilde{k}(n)}{\widetilde{\varepsilon}(n)}=o\left(\frac{n}{\log^L n}\right)$ and the sequence $\frac{\widetilde{k}(n)}{\widetilde{\varepsilon}(n)}$ has bounded consecutive ratios.*
2. *$\omega(\log n)\leqslant\frac{1}{\widetilde{\varepsilon}(n)}\leqslant o\left(\frac{1}{(\log n)^{2L}}\right)\widetilde{k}(n)$.*

Then for sufficiently large n, setting $\varepsilon=\widetilde{\varepsilon}(n)$ and $k=\widetilde{k}(n)$, k-means in ℓ_1 metric on n points in $\{0,1\}^d$ has no $(1+\varepsilon)$-approximation running in time $2^{(k/\varepsilon)/(\log(k/\varepsilon))^C}\operatorname{poly}(n,d)$.

We use the same point sets as in the preceding subsection to prove the corresponding lower bound for k-means in the ℓ_1 metric. The objective now sums the *squares* of the ℓ_1 distances, so the comparison for k-median does not immediately give the desired gap. In particular, an upper bound on the sum of the distances does not give an upper bound on the sum of their squares. For completeness, we therefore bound the distance of each point to a center obtained from the centroid

of its original Euclidean cluster, and then square these bounds. The shared color coordinate in every completeness cluster will allow us to control the additional terms introduced by squaring. For soundness, we use the lower bound for k-median from Lemma 1.4.

1.2.1 Construction

For every retained coloring ρ, start with the Euclidean points $P_\rho = \{p_1, \ldots, p_m\}$ from Section 3.2. Construct $Y_\rho = \{y_1, \ldots, y_m\}$ exactly as in (3), using $T = 8t$. Thus its coordinate $(g, a) \in \mathcal{G} \times \mathcal{A}$ is $(y_j)_{g,a} = q \cdot \big(\langle g, p_j \rangle + \xi_j(a)\big)$.

For a Euclidean center $\mu \in \mathbb{R}^{d_0}$, we will use the center $\Psi(\mu) \in \mathbb{R}^D$ whose coordinate is $\Psi(\mu)_{g,a} := q \cdot \langle g, \mu \rangle$ for every $(g, a) \in \mathcal{G} \times \mathcal{A}$. For each fixed g, all these center coordinates are equal, regardless of the shifts. Thus, after dividing by q, the one-dimensional center is $\langle g, \mu \rangle$, as in the discussion following Fact 1.2.

1.2.2 Analysis

Let $\Lambda := qD$. Since each coordinate in the construction is scaled by q and there are D coordinates in total, dividing an ℓ_1 distance by $\Lambda = qD$ cancels the factor q and averages the resulting one-dimensional distances over all coordinates (g, a). Let $s := 2t - 1$ and $\tau := T/2 + s/(2T)$. Recall that every cluster in the completeness case construction has a color c shared by all its points. If $u := \mathbf{e}_{N+c}$ denotes the corresponding unit vector, then $\|p_j - u\|_2^2 = s$ for every point in the cluster. Thus τ is the value given by Fact 1.2, after averaging over the Hadamard directions, when the squared Euclidean distance is equal to s.

Fix a cluster in the completeness case clustering with point indices $S \subseteq [m]$, let $P_S := \{p_j : j \in S\}$, and let μ be its centroid. We first bound the ℓ_1 distance from each transformed point y_j to the center $\Psi(\mu)$. By the definitions of y_j and $\Psi(\mu)$,

$$\frac{\|y_j - \Psi(\mu)\|_1}{\Lambda} = \frac{1}{D} \sum_{g \in \mathcal{G}} \sum_{a \in \mathcal{A}} \big|\langle g, p_j - \mu \rangle + \xi_j(a)\big| .$$

For a fixed g, as a ranges over $\mathcal{A}$, the shift $\xi_j(a)$ is uniform over the q midpoints of the equal subintervals of $[-T, T]$. Since the function $x \mapsto |\langle g, p_j - \mu \rangle + x|$ is convex, its average over these midpoints is at most its average under a continuous uniform shift on $[-T, T]$. Moreover, $|\langle g, p_j -$

$\mu\rangle| \leqslant \|p_j - \mu\|_1 \leqslant 2(s+1) \leqslant T$, so Fact 1.2 applies with $b = \langle g, p_j - \mu\rangle$. Hence,

$$\begin{aligned}
\frac{\|y_j - \Psi(\mu)\|_1}{\Lambda} &= \frac{1}{|\mathcal{G}|}\sum_{g\in\mathcal{G}}\frac{1}{|\mathcal{A}|}\sum_{a\in\mathcal{A}}\left|\langle g, p_j-\mu\rangle + \xi_j(a)\right| \\
&\leqslant \frac{1}{|\mathcal{G}|}\sum_{g\in\mathcal{G}}\frac{1}{2T}\int_{-T}^{T}\left|\langle g, p_j-\mu\rangle + x\right| dx \\
&= \frac{1}{|\mathcal{G}|}\sum_{g\in\mathcal{G}}\left(\frac{T}{2} + \frac{\langle g, p_j-\mu\rangle^2}{2T}\right) \\
&= \frac{T}{2} + \frac{\|p_j-\mu\|_2^2}{2T} \\
&= \tau + \frac{\|p_j-\mu\|_2^2 - s}{2T},
\end{aligned}$$

where the second equality follows from (2).

We now analyze what happens when this pointwise bound is squared. For each $j \in S$, define $e_j := \|p_j - \mu\|_2^2 - s$, the deviation of the squared Euclidean distance from the reference value s. The preceding inequality becomes $\frac{\|y_j-\Psi(\mu)\|_1}{\Lambda} \leqslant \tau + \frac{e_j}{2T}$, and therefore,

$$\frac{\|y_j - \Psi(\mu)\|_1^2}{\Lambda^2} \leqslant \tau^2 + \frac{\tau}{T}e_j + \frac{e_j^2}{4T^2}.$$

The linear term is directly related to the Euclidean clustering gain. Indeed, recall that all points in the cluster share the color c and that $u = \mathbf{e}_{N+c}$. Since $\|p_j - u\|_2^2 = s$ for every $j \in S$, using u as the Euclidean center costs $s|S|$. On the other hand, μ is the centroid of P_S, and hence its cost is $\mathrm{OPT}_1^{(2,2)}(P_S)$. We will show that $\sum_{j\in S} e_j = -\left(s|S| - \mathrm{OPT}_1^{(2,2)}(P_S)\right)$, so the linear term decreases the cost by τ/T times this gain. The quadratic term has the opposite effect. The shared coordinate will allow us to bound it by $\sum_{j\in S} e_j^2 \leqslant 4s\left(s|S| - \mathrm{OPT}_1^{(2,2)}(P_S)\right)$. Consequently, after squaring and summing over the points in the cluster, the linear term contributes a decrease of $\frac{\tau}{T}$ times the Euclidean cost gain, while the quadratic term can increase the cost by at most $\frac{s}{T^2}$ times the same gain. Thus the transformed squared ℓ_1 cost retains the Euclidean cost gain with coefficient $\frac{\tau}{T} - \frac{s}{T^2}$.

For the soundness analysis, we use the inequality $x^2 \geqslant 2\tau x - \tau^2$ for every $x \geqslant 0$. Applying it to each point's distance divided by Λ and summing gives a lower bound on the squared cost in terms of the sum of the distances. This sum is at least the optimal k-median cost, so we can apply Lemma 1.4 regardless of how the centers are chosen. The following lemma states both bounds.

Lemma 1.6. *For every $1 \leqslant k \leqslant m$, we have:*

$$\frac{\mathrm{OPT}_k^{(1,2)}(Y_\rho)}{\Lambda^2} \geqslant \tau^2 m - \frac{\tau}{T}\left(sm - \mathrm{OPT}_k^{(2,2)}(P_\rho)\right) - 14\tau T k. \tag{8}$$

Moreover, let $\mathcal{P}$ be a partition of $[m]$ into at most k nonempty sets. Suppose that, for every $S \in \mathcal{P}$, there exists a coordinate that is equal to one in every point p_j with $j \in S$. Then,

$$\frac{\mathrm{OPT}_k^{(1,2)}(Y_\rho)}{\Lambda^2} \leqslant \tau^2 m - \left(\frac{\tau}{T} - \frac{s}{T^2}\right)\left(sm - \sum_{S\in\mathcal{P}}\mathrm{OPT}_1^{(2,2)}(P_S)\right). \tag{9}$$

Finally, we convert the integer points to Boolean points using the same unary encoding described for k-median in ℓ_1 metric. We briefly verify that this also preserves the k-means optimum in ℓ_1 metric, even though the centers need not be integral. Consider any single coordinate, and say its input values lie in $\{0,\dots,B\}$. We may take the corresponding coordinate entry b of an optimal center to lie in $[0,B]$. Let $b=\ell+\theta$, where $\ell=\lfloor b\rfloor$ and $\theta\in[0,1)$. We associate with b a center in the corresponding B-dimensional Boolean block whose first ℓ coordinates are one, whose $(\ell+1)$st coordinate is θ when $\ell<B$, and whose remaining coordinates are zero. For every integer input value $a\in\{0,\dots,B\}$, its unary encoding has ℓ_1 distance exactly $|a-b|$ from this center. Applying this coordinate-wise shows that every center in the original space can be represented in the Boolean space while preserving its distance to every input point.

Conversely, consider a center $(z_u)_{u\in[B]}$ in one Boolean block. We may assume that $z_u\in[0,1]$ for every u, and define the corresponding original center value by $b:=\sum_{u\in[B]}z_u$. For every integer input value a, the triangle inequality gives $|a-b|=\left|\sum_{u\in[B]}\left(\mathbf{1}_{\{a\geqslant u\}}-z_u\right)\right|\leqslant\sum_{u\in[B]}\left|\mathbf{1}_{\{a\geqslant u\}}-z_u\right|$. Thus, every center in the Boolean space yields a center in the original space with no larger ℓ_1 distance to any input point. Applying these two transformations to all k centers shows that the unary encoding preserves the continuous ℓ_1 k-means optimum.

Before proving Lemma 1.6, we now first prove Theorem 1.5.

Proof of Theorem 1.5. Suppose that the asserted algorithm exists for a sufficiently large constant C. Take the hypergraph $\mathcal{H}$ from Theorem 2.2 in [KM26], and choose n,k,ε as in the proof of Theorem 3.1, with the constant B specified below. Let $k':=k-1$, let Δ denote the degree of $\mathcal{H}$, and let $M:=N\Delta/t$. The proof of Theorem 3.1 already verifies the assumptions of Theorem 3.2 with k' clusters and gives $k/\varepsilon=\Theta(N)$, $k'\geqslant B\varepsilon N/2$, and $M=o(n)$.

For each retained coloring ρ, apply the construction in (3) to P_ρ with $R=2t$ and $T=8t$. Let q be as fixed in the proof of Theorem 1.1, and set $\Lambda:=qD$, $s:=2t-1$, and $\tau:=T/2+s/(2T)$. If $m:=|P_\rho|=M-B_\rho$, then the number of discarded hyperedge occurrences satisfies $B_\rho=O_t(M/k')=o(k'\Delta)$, as proved in the preceding subsection. Also $k'=o(k'\Delta)$ because $\Delta=\omega(1)$. Since T and τ are constants, both the difference between the number of retained points m and the original number of hyperedges M, and the additive term $14\tau Tk'$, are negligible compared with the gap of order $k'\Delta$.

The number of retained points $m=M-B_\rho$ may depend on the coloring ρ. We therefore express the costs relative to the common baseline τ^2M. In the completeness case, there is a coloring whose clustering from completeness case of Euclidean k-means has a common color coordinate in every cluster and total Euclidean cost at most $sM-(1-o(1))k'\Delta$. Let $\mathcal{P}$ denote the corresponding partition of $[m]$. Its saving relative to sm, the reference value for the actual number of retained points, is at least:

$$sm-\sum_{S\in\mathcal{P}}\mathrm{OPT}_1^{(2,2)}(P_S)\geqslant s(M-B_\rho)-\big(sM-(1-o(1))k'\Delta\big)=(1-o(1))k'\Delta-sB_\rho.$$

Let $\lambda:=\tau/T-s/T^2$. For our choice of T, this is $1/2-s/(2T^2)>0$. Applying the upper bound

of Lemma 1.6 gives:

$$\begin{aligned}\frac{\mathrm{OPT}_{k'}^{(1,2)}(Y_\rho)}{\Lambda^2} &\leqslant \tau^2(M - B_\rho) - \lambda\big((1-o(1))k'\Delta - sB_\rho\big)\\ &= \tau^2 M - \lambda(1-o(1))k'\Delta - (\tau^2 - s\lambda)B_\rho\\ &\leqslant \tau^2 M - \lambda k'\Delta + o(k'\Delta).\end{aligned}$$

For the last inequality, we used $\tau^2 - s\lambda = T^2/4 + 3s^2/(4T^2) > 0$, so the term involving B_ρ can only decrease the upper bound.

In the soundness case, Theorem 3.2 gives $\mathrm{OPT}_{k'}^{(2,2)}(P_\rho) \geqslant sM - (1 - 1/(8t))k'\Delta$ for every coloring. Its saving relative to sm is therefore at most:

$$sm - \mathrm{OPT}_{k'}^{(2,2)}(P_\rho) \leqslant \left(1 - \frac{1}{8t}\right)k'\Delta - sB_\rho.$$

Applying the lower bound of Lemma 1.6, we obtain

$$\begin{aligned}\frac{\mathrm{OPT}_{k'}^{(1,2)}(Y_\rho)}{\Lambda^2} &\geqslant \tau^2(M - B_\rho) - \frac{\tau}{T}\left(\left(1 - \frac{1}{8t}\right)k'\Delta - sB_\rho\right) - 14\tau T k'\\ &= \tau^2 M - \frac{\tau}{T}\left(1 - \frac{1}{8t}\right)k'\Delta - (\tau^2 - s\tau/T)B_\rho - 14\tau T k'\\ &\geqslant \tau^2 M - \frac{\tau}{T}\left(1 - \frac{1}{8t}\right)k'\Delta - o(k'\Delta).\end{aligned}$$

The contribution of the last two terms is $o(k'\Delta)$, because both B_ρ and k' are $o(k'\Delta)$ and their coefficients are fixed constants.

Thus, the two bounds have the same additive term $\tau^2 M$. What matters is the difference between the coefficients of $k'\Delta$. Since $\tau/T > 1/2$, $s < 2t$, and $T = 8t$, we have

$$\lambda - \frac{\tau}{T}\left(1 - \frac{1}{8t}\right) = \frac{\tau}{8tT} - \frac{s}{T^2} > \frac{1}{16t} - \frac{2t}{64t^2} = \frac{1}{32t}.$$

Take N sufficiently large that each of the two $o(k'\Delta)$ terms above has magnitude at most $k'\Delta/(128t)$, and let $U := \tau^2 M - \lambda k'\Delta + k'\Delta/(128t)$. The completeness cost divided by Λ^2 is at most U for some coloring. For every coloring in soundness case, its cost divided by Λ^2 exceeds U by at least

$$\left(\lambda - \frac{\tau}{T}\left(1 - \frac{1}{8t}\right) - \frac{1}{64t}\right)k'\Delta > \frac{k'\Delta}{64t}.$$

To distinguish these cases using a $(1+\varepsilon)$-approximation, we need $\varepsilon U < k'\Delta/(64t)$. Our choices $s = 2t - 1$ and $T = 8t$ give $\lambda = 1/2 - s/(2T^2) > 1/(128t)$, so $U \leqslant \tau^2 M = \tau^2 N\Delta/t$. Since $k' \geqslant B\varepsilon N/2$, we have $\varepsilon U \leqslant \frac{\tau^2 \varepsilon N\Delta}{t}$ and $\frac{k'\Delta}{64t} \geqslant \frac{B\varepsilon N\Delta}{128t}$. Choosing $B > 128\tau^2$ therefore gives $(1+\varepsilon)U < U + k'\Delta/(64t)$.

It remains to give the asserted algorithm exactly n points and k centers. We use the same duplication and padding construction as in the proof of Theorem 1.1. Let $W := \lfloor (n-1)/m \rfloor$, take W copies of every point of Y_ρ, and add $n - Wm$ copies of $x := 4nA_0\mathbf{e}_{\mathrm{new}}$, where $A_0 := (T + 2t)\Lambda$.

As in that proof, every original point has ℓ_1 norm at most A_0. If a cluster contains both x and an original point y, then for any center c, $\|x-c\|_1^2+\|y-c\|_1^2 \geqslant \frac{\|x-y\|_1^2}{2} \geqslant 8n^2A_0^2$. On the other hand, placing one center at x and the remaining centers at the origin costs at most nA_0^2. Thus an optimal clustering separates the added copies from the original points, and if Y'_ρ denotes the padded point set, then $\mathrm{OPT}_k^{(1,2)}(Y'_\rho) = W \cdot \mathrm{OPT}_{k'}^{(1,2)}(Y_\rho)$.

Applying the unary encoding described above gives exactly n Boolean points with the same optimum and polynomial dimension. We run the asserted algorithm on the instance for every coloring, divide its returned cost by $W\Lambda^2$, and compare with $(1+\varepsilon)U$. In the completeness case, the returned cost is at most this threshold for at least one coloring, while in the soundness case every returned cost is strictly larger. The running-time analysis is the same as in the proof of Theorem 1.1. □

Proof of Lemma 1.6. We first prove the lower bound. Fix any nonempty set C of at most k centers for Y. For each $j \in [m]$, let $\delta_j := \min_{c\in C} \|y_j - c\|_1/\Lambda$ for its distance to the closest center, divided by Λ. Since $(\delta_j-\tau)^2 \geqslant 0$, we have $\delta_j^2 \geqslant 2\tau\delta_j - \tau^2$. Summing over all points gives

$$\sum_{j\in[m]} \delta_j^2 \geqslant 2\tau \sum_{j\in[m]} \delta_j - \tau^2 m \geqslant 2\tau \frac{\mathrm{OPT}_k^{(1,1)}(Y)}{\Lambda} - \tau^2 m.$$

The second inequality holds because the sum of the distances to the centers C is at least the optimal k-median cost. We can now apply Lemma 1.4 to this sum:

$$\begin{aligned}\sum_{j\in[m]} \delta_j^2 &\geqslant 2\tau\left(\frac{mT}{2} + \frac{\mathrm{OPT}_k^{(2,2)}(P)}{2T} - 7Tk\right) - \tau^2 m \\ &= \tau(T-\tau)m + \frac{\tau}{T}\mathrm{OPT}_k^{(2,2)}(P) - 14\tau Tk \\ &= \tau^2 m - \frac{\tau}{T}\left(sm - \mathrm{OPT}_k^{(2,2)}(P)\right) - 14\tau Tk.\end{aligned}$$

For the last equality, we used $2\tau = T + s/T$, which gives $\tau(T-\tau) = \tau^2 - \tau s/T$. The bound holds for every choice of C, so taking the minimum over the centers proves (8).

We next prove the upper bound. Fix a cluster $S \in \mathcal{P}$ and let $\mu := |S|^{-1}\sum_{j\in S} p_j$ be its centroid. We use the center $\Psi(\mu)$ defined above. From the construction of the points and this center, the distance of y_j, divided by Λ is $\|y_j - \Psi(\mu)\|_1/\Lambda = \frac{1}{D}\sum_{g\in\mathcal{G}}\sum_{a\in\mathcal{A}}\left|\langle g, p_j-\mu\rangle + \xi_j(a)\right|$. For every fixed j and g, the shift $\xi_j(a)$ takes each of the q midpoint values equally often as a ranges over $\mathcal{A}$. For every fixed j and g, the shift $\xi_j(a)$ takes each of the q midpoint values equally often as a ranges over $\mathcal{A}$. Hence,

$$\frac{1}{q^4}\sum_{a\in\mathcal{A}}\left|\langle g, p_j-\mu\rangle + \xi_j(a)\right| = \frac{1}{q}\sum_{u\in[q]}\left|\langle g, p_j-\mu\rangle + \frac{T}{q}(2u-1-q)\right|.$$

On each of the q subintervals, the value of the convex function $x \mapsto |z+x|$ at the midpoint is no larger than its average on that subinterval. Summing this inequality over the subintervals gives:

$$\frac{1}{q^4}\sum_{a\in\mathcal{A}}|z+\xi_j(a)| \leqslant \frac{1}{2T}\int_{-T}^{T}|z+x|\,dx = \frac{T}{2} + \frac{z^2}{2T}.$$

Here the last equality follows from Fact 1.2, because $|z| \leqslant \|p_j - \mu\|_1 \leqslant 2(s+1) \leqslant T$. Averaging over $g \in \mathcal{G}$ and using (2) now gives the following:

$$\frac{\|y_j - \Psi(\mu)\|_1}{\Lambda} \leqslant \frac{T}{2} + \frac{1}{2T|\mathcal{G}|} \sum_{g \in \mathcal{G}} \langle g, p_j - \mu \rangle^2 = \frac{T}{2} + \frac{\|p_j - \mu\|_2^2}{2T}. \tag{10}$$

To bound the squared ℓ_1 cost, we will square the pointwise bound in (10). We therefore need to control both $\sum_{j \in S}(\|p_j - \mu\|_2^2 - s)$ and $\sum_{j \in S}(\|p_j - \mu\|_2^2 - s)^2$ (recall that $s = 2t - 1$). The coordinate shared by all points in the cluster gives the following bounds.

Claim 1.7. *Let S be a cluster whose points share the color c, let μ be the centroid of P_S, and define $e_j := \|p_j - \mu\|_2^2 - s$ for every $j \in S$. Then $\sum_{j \in S} e_j = -\left(s|S| - \mathrm{OPT}_1^{(2,2)}(P_S)\right)$ and $\sum_{j \in S} e_j^2 \leqslant 4s\left(s|S| - \mathrm{OPT}_1^{(2,2)}(P_S)\right)$.*

Proof. Let $u := \mathbf{e}_{N+c}$ and $v := \mu - u$. Every $p_j - u$ has exactly s coordinates equal to one and all other coordinates zero. In particular, $\|p_j - u\|_2^2 = s$. Also, v is the average of the vectors $p_j - u$, so all its coordinates lie in $[0,1]$. It follows that $0 \leqslant \langle p_j - u, v \rangle \leqslant s$ for every $j \in S$. Moreover, $\sum_{j \in S} \langle p_j - u, v \rangle = \left\langle \sum_{j \in S}(p_j - u), v \right\rangle = |S|\|v\|_2^2$. Expanding the squared distance around u and summing over $j \in S$ gives us:

$$\sum_{j \in S} e_j = \sum_{j \in S} \left(\|(p_j - u) - v\|_2^2 - s\right) = \sum_{j \in S} \left(\|v\|_2^2 - 2\langle p_j - u, v \rangle\right) = -|S|\|v\|_2^2.$$

For the squared terms, since $0 \leqslant \langle p_j - u, v \rangle \leqslant s$, we have $\langle p_j - u, v \rangle^2 \leqslant s\langle p_j - u, v \rangle$. Hence,

$$\begin{aligned}
\sum_{j \in S} e_j^2 &= |S|\|v\|_2^4 - 4\|v\|_2^2 \sum_{j \in S} \langle p_j - u, v \rangle + 4 \sum_{j \in S} \langle p_j - u, v \rangle^2 \\
&= 4 \sum_{j \in S} \langle p_j - u, v \rangle^2 - 3|S|\|v\|_2^4 \\
&\leqslant 4s \sum_{j \in S} \langle p_j - u, v \rangle = 4s|S|\|v\|_2^2.
\end{aligned}$$

Finally, the centroid identity (Fact 3.5) gives us $|S|\|v\|_2^2 = \sum_{j \in S} \|p_j - u\|_2^2 - \sum_{j \in S} \|p_j - \mu\|_2^2 = s|S| - \mathrm{OPT}_1^{(2,2)}(P_S)$. Substituting this identity into the two bounds above proves the claim. □

We can now square (10). By the definitions of τ and e_j, its right-hand side can be written as $\frac{T}{2} + \frac{\|p_j - \mu\|_2^2}{2T} = \tau + \frac{e_j}{2T} > 0$. Summing over the cluster and applying Claim 1.7, we obtain:

$$\begin{aligned}
\sum_{j \in S} \frac{\|y_j - \Psi(\mu)\|_1^2}{\Lambda^2} &\leqslant \sum_{j \in S} \left(\tau + \frac{e_j}{2T}\right)^2 \\
&= \tau^2|S| + \frac{\tau}{T} \sum_{j \in S} e_j + \frac{1}{4T^2} \sum_{j \in S} e_j^2 \\
&\leqslant \tau^2|S| - \frac{\tau}{T}\left(s|S| - \mathrm{OPT}_1^{(2,2)}(P_S)\right) + \frac{s}{T^2}\left(s|S| - \mathrm{OPT}_1^{(2,2)}(P_S)\right) \\
&= \tau^2|S| - \left(\frac{\tau}{T} - \frac{s}{T^2}\right)\left(s|S| - \mathrm{OPT}_1^{(2,2)}(P_S)\right).
\end{aligned}$$

Using the center $\Psi(\mu)$ for each $S \in \mathcal{P}$ and summing the above bound over $S \in \mathcal{P}$ therefore proves (9). □

2 Conditional Lower Bounds for Exact Algorithms under ETH

We now use the reductions in the preceding sections to prove lower bounds for exact algorithms in terms of the number of clusters k and the dimension d. For the ℓ_1 metric, refining the construction from Section 1 yields lower bounds with exponent $k^{1-\alpha}d^{(1+\alpha)/4}$ for every constant $\alpha \in [0,1)$, up to a polylogarithmic factor, and under randomized ETH this lower bound can be further improved to $k^{1-\alpha}d^{(1+\alpha)/2}$; see Section 2.1 for details.

Throughout this section, let t, L be the constants supplied by Theorem 2.2 in [KM26] with $\varepsilon = 1/100$. The hypergraphs in that theorem have N vertices, degree $\Delta = \Theta((\log N)^L)$, and $M = N\Delta/t$ hyperedges. Let C_H denote the constant in its running-time lower bound. We use n for the number of points and d for their dimension.

2.1 Lower Bounds for k-median and k-means in the ℓ_1 Metric

For the ℓ_1 metric, we cannot directly apply a dimension reduction as in the Euclidean setting. Instead, we refine the construction from Section 1 itself, replacing the full Hadamard family and the full family of shifts by smaller pseudorandom families while preserving the completeness–soundness gap. This reduces the dimension sufficiently to obtain the exact lower bounds below.

Theorem 2.1. *Assuming ETH, there is a constant $C > 0$ such that the following holds for every constant $\alpha \in [0,1]$. Let $\widetilde{d}\colon \mathbb{N} \to \mathbb{N}$, $\widetilde{k}\colon \mathbb{N} \to \mathbb{N}$ be polynomial time computable functions of the input size satisfying:*

1. *$\widetilde{k}(n)^{1-\alpha}\widetilde{d}(n)^{(1+\alpha)/4} = o(n/(\log n)^C)$ and the sequence $\widetilde{k}(n)^{1-\alpha} \cdot \widetilde{d}(n)^{(1+\alpha)/4}$ has bounded consecutive ratios.*
2. *$\omega\left((\log n)^C\right)\widetilde{k}(n)^{4\alpha/(1+\alpha)} \leqslant \widetilde{d}(n) \leqslant o\left(\frac{1}{(\log n)^C}\right)\widetilde{k}(n)^4$.*

Then for sufficiently large n, setting $d = \widetilde{d}(n)$ and $k = \widetilde{k}(n)$, both k-means and k-median in ℓ_1 metric on n points in $\mathbb{R}^d$ has no exact algorithm running in time $n^{k^{1-\alpha}d^{(1+\alpha)/4}/(\log n)^C}$.

Proof. Fix a constant $\alpha \in [0,1)$. Suppose that an algorithm $\mathcal{A}$ as in the theorem exists for a sufficiently large constant C. Let $\mathcal{H}$ be a hypergraph from Theorem 2.2 in [KM26], let $a := L + 5$, and choose the least integer n such that $\widetilde{k}(n)^{1-\alpha}\widetilde{d}(n)^{(1+\alpha)/4} \geqslant N(\log n)^a$. Let $k := \widetilde{k}(n)$, $d := \widetilde{d}(n)$, and $k' := k - 1$. Minimality of n and bounded consecutive ratios, applied to $\widetilde{k}(n)^{1-\alpha}\widetilde{d}(n)^{(1+\alpha)/4}$, gives $k^{1-\alpha}d^{(1+\alpha)/4} = \Theta(N(\log n)^a)$.

We first verify the assumptions of Theorem 3.2. The upper bound on d gives $\frac{(k')^2}{N} = \Theta\left(\left(\frac{k^4}{d}\right)^{(1+\alpha)/4}(\log n)^a\right) = \omega\big((\log n)^{C/2+a}\big)$. Since $\Delta = O((\log n)^L)$, taking C sufficiently large gives $\Delta\sqrt{N} = o(k')$. Similarly, the lower bound on d gives $\frac{k'}{N} = \Theta\left(\frac{k^\alpha(\log n)^a}{d^{(1+\alpha)/4}}\right) = o\big((\log n)^{a-C/2}\big) = o(1/\log N)$. The codegree bound also gives $\chi \log k' = o(\Delta)$. Thus Theorem 3.2 applies with k' clusters. Moreover, $k^{1-\alpha}d^{(1+\alpha)/4} = o(n/(\log n)^C)$ implies $M = N\Delta/t = o(n)$.

We would now like to reduce the dimension of the resulting ℓ_1 instances, as we did in the Euclidean setting. However, the Johnson–Lindenstrauss argument used there does not apply to the ℓ_1 metric. Instead, we reduce the dimension within the construction itself, replacing the full Hadamard family and the full family of shifts from Section 1 by smaller pseudorandom families while preserving the completeness–soundness gap.

Construction. Let $d_0 := N + k'$, let $T := 8t$, and let q be the least power of two at least M. Fix a sufficiently small constant $\eta > 0$, to be chosen after the analysis, and set $\gamma := \eta k'/N$ and $\varepsilon := \gamma/\log(2M)$.

We use the explicit small-bias set construction of [AGHP92]. For every $r \in \mathbb{N}$ and $\beta \in (0,1)$, one can construct a set $\mathcal{S} \subseteq \{0,1\}^r$ of size $O(r^2/\beta^2)$ such that for every nonzero $a \in \{0,1\}^r$, $\left|\frac{1}{|\mathcal{S}|}\sum_{w\in\mathcal{S}}(-1)^{\langle a,w\rangle}\right| \leqslant \beta$, where the inner product is over $\mathbb{F}_2$. Applying this construction with $r = \lceil \log_2(d_0+1)\rceil$ gives an explicit multiset $\mathcal{G} \subseteq \{-1,1\}^{d_0}$ of size $h = O(\gamma^{-2}(\log(2d_0))^2)$ such that $|\mathbb{E}_{g\in\mathcal{G}}[g(u)g(v)]| \leqslant \gamma$ for every distinct $u, v \in [d_0]$. Consequently, for every $x \in \mathbb{R}^{d_0}$,

$$\left|\frac{1}{h}\sum_{g\in\mathcal{G}}\langle g,x\rangle^2 - \|x\|_2^2\right| \leqslant \gamma \sum_{\substack{u,v\in[d_0]\\ u\neq v}} |x_u x_v| \leqslant \gamma\|x\|_1^2. \tag{11}$$

We also need an approximate version of the four-wise independent family used in Section 1, i.e., we need an explicit multiset $\mathcal{B}$ of functions $b : [M] \to [q]$ of size $O(\varepsilon^{-2}(\log(2q))^{10})$ such that for every $r \in \{1,2,3,4\}$, every distinct $j_1,\dots,j_r \in [M]$, and every $u_1,\dots,u_r \in \{0,\dots,q\}$, we have:

$$\left|\Pr_{b\in\mathcal{B}}\left[b(j_i) \leqslant u_i \text{ for every } i \in [r]\right] - \prod_{i\in[r]}\frac{u_i}{q}\right| \leqslant \varepsilon. \tag{12}$$

This can be obtained by starting with the standard degree-three polynomial construction of four-wise independent functions over $\mathbb{F}_q$ (see, e.g., [Vad12]) and replacing its uniform seed by a β-biased sample space from [AGHP92], where $\beta := \varepsilon/(\log_2 q + 1)^4$.

For every retained coloring ρ, list $P_\rho = \{p_1,\dots,p_m\}$ and define $Y_\rho = \{y_1,\dots,y_m\}$ with coordinates indexed by $(g,b) \in \mathcal{G}\times\mathcal{B}$:

$$(y_j)_{g,b} := q\langle g, p_j\rangle + T(2b(j) - 1 - q). \tag{13}$$

Equivalently, $(y_j)_{g,b} = q(\langle g,p_j\rangle + \xi_j(b))$, where $\xi_j(b) := T(2b(j)-1-q)/q$. Let $D := h|\mathcal{B}|$ and $\Lambda := qD$. The families $\mathcal{G},\mathcal{B}$, and hence D, are the same for every coloring. Every coordinate is an integer of absolute value at most $(2t+T)q$.

Analysis. We first prove the analogue of Lemma 1.3 for the shifts in (13).

Claim 2.2. *Let $I \subseteq [M]$ be nonempty, let $r := |I|$, and let $z_j \in [-2t,2t]$ for every $j \in I$. Let $\mu := r^{-1}\sum_{j\in I} z_j$. Then*

$$\left|\mathbb{E}_{b\in\mathcal{B}}\left[\min_{c\in\mathbb{R}}\sum_{j\in I}|z_j+\xi_j(b)-c|\right] - \left(\frac{rT}{2} + \frac{1}{2T}\sum_{j\in I}(z_j-\mu)^2\right)\right| \leqslant C_t\left(1+\left(\varepsilon+\frac{1}{q}\right)r\right)\log(2r),$$

where C_t depends only on t.

Proof. Choose b uniformly from $\mathcal{B}$, let $\widetilde{z}_j := z_j + \xi_j(b)$, and define $F(c) := \sum_{j\in I} |\widetilde{z}_j - c|$ and $G(c) := \sum_{j\in I}(2\mathbf{1}_{\{\widetilde{z}_j \leqslant c\}} - 1)$. Let U be uniform on $[-T, T]$, and let $\nu_j(c) := \Pr[z_j + U \leqslant c]$ and $g_0(c) := \sum_{j\in I}(2\nu_j(c) - 1)$. For every $x \in \mathbb{R}$, the probability that a uniformly chosen midpoint is at most x differs from $\Pr[U \leqslant x]$ by at most $1/q$. Thus (12), with $r = 1, 2$, gives both $\left|\mathbb{E}\mathbf{1}_{\{\widetilde{z}_j \leqslant c\}} - \nu_j(c)\right| \leqslant \varepsilon + \frac{1}{q}$ and $\left|\mathbb{E}\left[\mathbf{1}_{\{\widetilde{z}_i \leqslant c\}}\mathbf{1}_{\{\widetilde{z}_j \leqslant c\}}\right] - \nu_i(c)\nu_j(c)\right| \leqslant \varepsilon + \frac{2}{q}$ for distinct $i, j \in I$. Let $\delta := \varepsilon + 2/q$. Let $\delta := \varepsilon + 2/q$. Expanding $\mathbb{E}[(G(c) - g_0(c))^2]$ and applying the two bounds above gives us:

$$\mathbb{E}\left[(G(c) - g_0(c))^2\right] \leqslant 4r + 12\delta r^2 \qquad \text{for every } c \in \mathbb{R}. \tag{14}$$

The marginal bound also compares the cost at μ with its continuous uniform-shift average. Since $|z_j - \mu| \leqslant 4t = T/2$, we may apply Fact 1.2 with $b = z_j - \mu$. As in the proof of Lemma 1.3, integrating the difference of the distribution functions gives us:

$$\left|\mathbb{E}F(\mu) - \left(\frac{rT}{2} + \frac{1}{2T}\sum_{j\in I}(z_j - \mu)^2\right)\right| \leqslant 2T\delta r. \tag{15}$$

It remains to bound the saving from moving the center from μ to a median. We use the same integral representation as in the proof of Lemma 1.3, i.e., $F(\mu) - \min_{c\in\mathbb{R}} F(c) = \int_0^\infty (-G(\mu + x))_+ \, dx + \int_0^\infty (G(\mu - x))_+ \, dx$. For $0 < x \leqslant T/2$, we have $g_0(\mu + x) = rx/T$. Using $(-w)_+ \leqslant (w - v)^2/(4v)$ for $v > 0$ together with (14) gives us:

$$\mathbb{E}[(-G(\mu + x))_+] \leqslant \min\left\{r, \frac{CT(1 + \delta r)}{x}\right\}$$

for an absolute constant C. Integrating over $[0, T/2]$ gives us $O(T(1 + \delta r)\log(2r))$. For $x \geqslant T/2$, the same argument and the monotonicity of g_0 give an additional $O(T(1 + \delta r))$ contribution, and the integrand vanishes for $x \geqslant 3T/2$. The argument for movement to the left is identical. Combining these bounds with (15) proves the claim. □

We now repeat the analyses from Section 1, using (11) in place of exact Hadamard orthogonality and Claim 2.2 in place of Lemma 1.3. Applying Claim 2.2 to each cluster and each $g \in \mathcal{G}$, and using (11), the proof of Lemma 1.4 gives us:

$$\left|\frac{\mathrm{OPT}^{(1,1)}_{k'}(Y_\rho)}{\Lambda} - \left(\frac{mT}{2} + \frac{\mathrm{OPT}^{(2,2)}_{k'}(P_\rho)}{2T}\right)\right| \leqslant O_t(\eta k'\Delta) + o(k'\Delta). \tag{16}$$

Here we used the choices of ε, γ, the bound $M/q \leqslant 1$, and $\log M = o(\Delta)$.

For k-means, let $s := 2t - 1$ and $\tau := T/2 + s/(2T)$. Following the proof of Lemma 1.6, using the same centroid centers and Claim 1.7, gives for the completeness partition $\mathcal{P}$ the following:

$$\frac{\mathrm{OPT}^{(1,2)}_{k'}(Y_\rho)}{\Lambda^2} \leqslant \tau^2 m - \left(\frac{\tau}{T} - \frac{s}{T^2}\right)\left(sm - \sum_{S\in\mathcal{P}} \mathrm{OPT}^{(2,2)}_1(P_S)\right) + O_t(\eta k'\Delta). \tag{17}$$

On the other hand, applying $x^2 \geqslant 2\tau x - \tau^2$ together with (16), as in the soundness case of the

same lemma, gives us:

$$\frac{\mathrm{OPT}_{k'}^{(1,2)}(Y_\rho)}{\Lambda^2} \geqslant \tau^2 m - \frac{\tau}{T}\left(sm - \mathrm{OPT}_{k'}^{(2,2)}(P_\rho)\right) - O_t(\eta k'\Delta) - o(k'\Delta). \tag{18}$$

Finally, as in the proofs of Theorems 1.1 and 1.5, we replace the coloring-dependent value $m = M - B_\rho$ by the common baseline M. Since $B_\rho = o(k'\Delta)$, this changes the bounds only by $o(k'\Delta)$. The gap between the completeness and soundness cases in Euclidean metric therefore remains of order $k'\Delta$ for k-median. For k-means, the difference between the two relevant coefficients is $\left(\frac{\tau}{T} - \frac{s}{T^2}\right) - \frac{\tau}{T}\left(1 - \frac{1}{8t}\right) > \frac{1}{32t}$. Thus, choosing η sufficiently small preserves a gap of order $k'\Delta$ for both objectives.

It remains to bound the dimension and the running time. The sizes of the two families implies,

$$D = h|\mathcal{B}| = O\left(\gamma^{-2}\varepsilon^{-2}(\log(2M))^{12}\right) = O_t\left(\left(\frac{N}{k'}\right)^4 (\log N)^{14}\right) = O_t\left(d\left(\frac{d}{k^4}\right)^\alpha \frac{(\log N)^{14}}{(\log n)^{4a}}\right) = o(d).$$

The last equality follows from $a = L + 5$, $\log N = O(\log n)$, and the upper bound on d. Both families and all transformed points can be constructed deterministically in polynomial time in N.

Every transformed point has ℓ_1 norm at most $A := (2t + T)qD$. Apply the same duplication and padding argument as in previous section, and append zero coordinates to obtain exactly n points in dimension d with k centers.

Run $\mathcal{A}$ on every padded instance and divide its returned cost by the duplication factor and by Λ for k-median or Λ^2 for k-means. The preserved gap distinguishes the two hypergraph cases. Since $k^{1-\alpha}d^{(1+\alpha)/4} = \Theta(N(\log n)^a)$, the running time on each instance is $n^{k^{1-\alpha}d^{(1+\alpha)/4}/(\log n)^C} = 2^{O(N/(\log n)^{C-a-1})} \leqslant 2^{O(N/(\log N)^{C-a-1})}$. Choosing $C > C_H + a + 1$ makes this $2^{o(N/(\log N)^{C_H})}$.

Finally, the lower and upper bounds on d imply $k = \omega((\log n)^C)$, and the lower bound on d then gives $k^{1-\alpha}d^{(1+\alpha)/4} = \omega((\log n)^{3C/2})$. Hence $N = \omega((\log n)^{3C/2-a})$. The parameter assumptions also give $d \leqslant n^4$ for all sufficiently large n. The preceding bounds imply that all polynomial factors in n and d contribute only $2^{o(N/(\log N)^{C_H})}$ to the running time for sufficiently large C. Together with the $2^{O((\log N)^2)}$ coloring enumeration, the total running time contradicts Theorem 2.2 in [KM26]. □

Remark 2.3. *The weaker dependence on k and d in Theorem 2.1, compared with the Euclidean lower bound in Corollary 1.3 in [KM26], comes from the dimension blow-up incurred by the pseudorandom families used in the deterministic reduction. Under randomized ETH, this can be avoided by sampling the coordinates directly, and the same dependence on k and d as in the Euclidean case can be recovered.*

We briefly explain this for k-median. Instead of taking one coordinate for every pair from the two pseudorandom families, sample $D = O((N/k)^2(\log N)^2)$ independent sign vectors and independent midpoint shifts, and define each coordinate as in the construction of Section 1. The sampled sign vectors preserve the relevant Euclidean squared costs, while the expected ℓ_1 median cost is related to these costs by the same random-shift argument as in Lemma 1.4. A bounded-differences concentration bound, followed by a union bound over all clusterings, shows that the completeness–soundness gap is preserved with high probability. The k-means case follows similarly, using the analysis of Lemma 1.6.

Consequently, assuming randomized ETH, the conclusion of Corollary 1.3 in [KM26] also holds for k-median and k-means in the ℓ_1 metric.

3 Lower Bounds for Euclidean k-means under ETH

In this section we improve on the range of parameters of [KM26] for Euclidean k-means and prove the following. Fix $\varepsilon = \frac{1}{100}$ in Theorem 2.2 in [KM26], let s be from that theorem and set $L = 3(s-1)$ throughout this section.

Theorem 3.1. *Assuming ETH, there is a constant $C > 0$ such that the following holds. Let $\widetilde{\varepsilon}\colon \mathbb{N} \to (0,1)$, $\widetilde{k}\colon \mathbb{N} \to \mathbb{N}$ be polynomial time computable functions satisfying:*

1. *$(\log n)^2 \leqslant \frac{\widetilde{k}(n)}{\widetilde{\varepsilon}(n)} = o\left(\frac{n}{\log^L n}\right)$, and the sequence $\frac{\widetilde{k}(n)}{\widetilde{\varepsilon}(n)}$ has bounded consecutive ratios.*
2. *$\widetilde{k}(n) = \omega(1)$ and $\widetilde{\varepsilon}(n) = o(1/\log n)$.*

Then for sufficiently large n, setting $\varepsilon = \widetilde{\varepsilon}(n)$ and $k = \widetilde{k}(n)$, Euclidean k-means on n points in $\mathbb{R}^d$ has no $(1+\varepsilon)$-approximation algorithm running in time $2^{(k/\varepsilon)/(\log(k/\varepsilon))^C}\operatorname{poly}(n,d)$.

The proof of Theorem 3.1 is based on a reduction from hypergraph coverage as follows.

Theorem 3.2. *Let $\mathcal{H} = (V, \mathcal{E})$ be a t-uniform d-regular hypergraph on N vertices where $d = d(N)$ grows with N, let k be an integer satisfying $\omega(1) \leqslant k \leqslant o(N/\log N)$, and suppose that the codegree of $\mathcal{H}$ is at most $o(d/\log N)$. Then there are integers $s \leqslant N^2$ and $R \geqslant 0$, and an algorithm running in time $2^{O(\log^2 N)}$ that transforms $\mathcal{H}$ into a collection $\mathcal{P}$ of at most $2^{O(\log^2 N)}$ subsets of $\{0,1,\dots,N\}^{N+k}$ of size at most $(1+o(1))M$ each, where $M = Nd/t$, such that the following holds.*

1. ***Completeness:** If there is $S \subseteq V$ of size N/t that touches each hyperedge of $\mathcal{H}$ once, then there is $P \in \mathcal{P}$ such that*
$$\mathrm{OPT}_k^{(2,2)}(P) \leqslant (ts+t-1)(|P|-R) - (1-o(1))skd.$$
2. ***Soundness:** If every $S \subseteq V$ of size N/t fails to cover at least $M/4$ of the hyperedges of $\mathcal{H}$, then for all $P \in \mathcal{P}$ we have*
$$\mathrm{OPT}_k^{(2,2)}(P) \geqslant (ts+t-1)(|P|-R) - \left(1-\frac{1}{8t}\right)skd.$$

We first show that Theorem 3.2 implies Theorem 3.1. The point of the reduction is that in Theorem 3.2, the gap between the YES case and the NO case is an additive factor of $\Theta\left(\frac{skd}{t}\right)$ while the cost itself is $\Theta(tsM)$, and we want to turn it into a multiplicative factor as required in Theorem 3.1. Once $\frac{k}{\varepsilon} \geqslant B \cdot tN$ for sufficiently large constant B, the additive gap already gives the necessary multiplicative gap, and the main idea is to pick $q \geqslant N$ where $\frac{\widetilde{k}(q)}{\widetilde{\varepsilon}(q)} \geqslant B \cdot tN$, and then use a duplication trick to increase the number of points to q.

Proof of Theorem 3.1. Let $H = (V, E)$ be an instance from Theorem 2.2 in [KM26] with $\varepsilon = 1/100$, and write $N = |V|$. Let t and d denote its uniformity and degree, respectively, and let $M = Nd/t$. Let B be a sufficiently large constant and fix q to be the first integer such that $\frac{\widetilde{k}(q)}{\widetilde{\varepsilon}(q)} \geqslant BtN$. We note that as $\widetilde{k}(N)/\widetilde{\varepsilon}(N) = o(N)$ we get that $q \geqslant N$, and by the bounded consecutive ratio assumption we have $\frac{\widetilde{k}(q)}{\widetilde{\varepsilon}(q)} = \Theta(N)$.

Setting $k = \widetilde{k}(q) - 1$ and $\varepsilon = \widetilde{\varepsilon}(q)$, we get that $\mathcal{H}$ satisfies the assumptions of Theorem 3.2, and applying it we get integers s, R and a collection $\mathcal{P}$ of $2^{O(\log^2 N)}$ subsets as therein. Each $P \in \mathcal{P}$ has size at most $M = \frac{Nd}{t}$, and we turn it into a P' of size q as follows. Note that

$$|P| \leqslant (1+o(1))M = (1+o(1))\frac{Nd}{t} \leqslant O(N \log^L N) \leqslant O\left(\frac{\widetilde{k}(q)\log^L N}{\widetilde{\varepsilon}(q)}\right) = o\left(\frac{q \log^L N}{\log^L q}\right) = o(q).$$

Thus, letting $W := \lfloor (q-1)/|P| \rfloor$, we take W copies of every point of P, and append one zero coordinate to these points. Finally, we add $q - W|P|$ copies of $x := 4tq\mathbf{e}_{\text{new}}$.

We note that for each P it holds that $\mathrm{OPT}^{2,2}_{\widetilde{k}(q)}(P') = W\,\mathrm{OPT}^{2,2}_{k}(P)$. Indeed, any cluster containing both x and an original point p has cost at least $\|x-p\|_2^2/2 > 2N^3q$, whereas placing all copies of x in one cluster and all original points at the origin has cost at most $2N^3q$, since $P \subseteq \{0,1,\ldots,N\}^{N+k}$ and $k \leqslant N$. Thus, in an optimal solution one center must be devoted to the copies of x (hence they contribute 0), while the remaining k centers cluster the duplicated copy of P.

Denoting $S = (ts+t-1)(|P|-R) - \left(1-\frac{1}{8t}\right)skd$, in the NO case we get that $\mathrm{OPT}^{2,2}_{\widetilde{k}(q)}(P') \geqslant WS$ for all P', and in the YES case we get that there is P' such that

$$\mathrm{OPT}^{2,2}_{\widetilde{k}(q)}(P') \leqslant WS - \left(\frac{1}{8} - o(1)\right) W\frac{skd}{t} \leqslant WS - \frac{1}{32}W\frac{BstNd}{t}\varepsilon = WS - W\frac{B}{32}tsM\varepsilon \leqslant (1-\varepsilon)WS$$

for sufficiently large B. Thus, if we an algorithm solving the problem in Theorem 3.1 running in time $2^{(k/\varepsilon)/(\log(k/\varepsilon))^C}$ for a large C, we could run it on each P' and accept only if one of the runs returned value at most WS. This would have been an algorithm running in time $2^{(k/\varepsilon)/(\log(k/\varepsilon))^C} \cdot 2^{O(\log^2 N)}$ that distinguishes between the YES and NO case of Theorem 2.2 in [KM26], which contradicts ETH for sufficiently large C. □

In the rest of this section we prove Theorem 3.2. In Section 3.1, we provide an explicit family of colorings of hypergraphs which have less constraints on parameters but also give weaker conclusions than [KM26], and in Section 3.2 we construct the point sets and give the completeness analysis, and in Section 3.3 we analyze the soundness.

3.1 An Explicit Family of Colorings

In this section, we show how to construct a family of vertex coloring of hypergraphs in quasipolynomial time with certain nice properties that are useful for the reduction to clustering problems in the subsequent sections. We will use the following standard concentration bound for sums of bounded-independence random variables.

Proposition 3.3. *Let $r \geqslant 2$ be even, and let $X_1, \ldots, X_N \in [0, B]$ be r-wise independent random variables. Let $X = \sum_{i\in[N]} X_i$ and $\sigma^2 = \sum_{i\in[N]} \mathrm{Var}(X_i)$. There is an absolute constant C such that*

$$\mathbb{E}\left[|X - \mathbb{E}[X]|^r\right] \leqslant C^r\left(\sqrt{r\sigma^2} + rB\right)^r \quad \text{and} \quad \Pr\left[|X - \mathbb{E}[X]| > 4C(\sqrt{r\sigma^2} + rB)\right] \leqslant 4^{-r}.$$

Proof. Expanding $(X - \mathbb{E}[X])^r$ gives terms involving at most r distinct variables. By r-wise independence, its expectation is therefore the same as for fully independent variables with the same

marginals. The first bound follows from the moment form of Bernstein's inequality, and the second from Markov's inequality. □

Theorem 3.4. *Fix an integer $t \geqslant 2$, and let n, k, d, χ be sufficiently large integers such that $\omega(1) \leqslant k \leqslant o(n/\log n)$, and $\chi \log n = o(d)$. Then, in time $2^{O((\log n)^2)}$ one can construct a list $\mathcal{C}_{n,k}$ of k-colorings of $[n]$ with the following property. If $\mathcal{H}$ is a t-uniform hypergraph over $[n]$ with degree d and codegree χ, then there exists $\rho \in \mathcal{C}_{n,k}$ with the following properties:*

1. ***Nearly balanced colors:*** *For every $c \in [k]$, we have $\left||\rho^{-1}(c)| - \frac{n}{k}\right| = o(n/k)$.*
2. ***Nearly all hyperedges rainbow:*** *At most $100\binom{t}{2}\frac{nd}{tk}$ hyperedges have two vertices of the same color.*
3. ***Neighbors of one color:*** *For every $v \in [n]$ and $c \in [k]$, there are at most $o(d)$ number of hyperedges containing v together with other vertices of color c.*
4. ***Hyperedges using two colors:*** *For every pair of distinct colors $a, b \in [k]$, at most $O_t\left(\frac{nd}{k^2} + d\log n\right)$ rainbow hyperedges use both colors.*
5. ***Balanced with Weights:*** *For any fixed set of numbers $\{q_v \in [0,d] \mid v \in [n]\}$ satisfying $\sum_{v\in[n]} q_v = nd/t$, the coloring can be chosen to satisfy the above properties as well as*

$$\left|\sum_{v:\rho(v)=c} q_v - \frac{nd}{tk}\right| = o\left(\frac{nd}{tk}\right)$$

for every $c \in [k]$.

Proof. Setting $r = 20\lceil \log_2 n\rceil$, we take $\mathcal{C}_{n,k}$ a tr-wise independent family of functions $\rho : [n] \to [k]$ with uniform marginals; the explicit construction of [ABI86] achieves near uniform marginals (and would suffice for us), but it is more convenient to use the construction of [HS24] that achieves perfectly uniform marginals. Both have size $n^{O(tr)} = 2^{O((\log n)^2)}$.

Fixing a hypergraph $\mathcal{H}$, we prove that a randomly chosen $\rho \in \mathcal{C}_{n,k}$ satisfies all five properties with probability close to 1. Using Proposition 3.3 with the constant C from there and the union bound, the probability that

$$\left||\rho^{-1}(c)| - \frac{n}{k}\right| > 4C\left(\sqrt{\frac{rn}{k}} + r\right) = o(n/k)$$

for some $c \in [k]$ is at most $k \cdot 4^{-r} \leqslant \frac{1}{100}$.

For the second item, for each hyperedge e let Y_e be the indicator random variable that it is not rainbow under ρ. Then by pairwise independence $\mathbb{E}[Y_e] \leqslant \binom{t}{2}\frac{1}{k}$, so, using $|E| = nd/t$, $\mathbb{E}\left[\sum_{e\in E} Y_e\right] \leqslant \binom{t}{2}\frac{nd}{tk}$. Markov's inequality now gives that the second item fails with probability at most $1/100$.

For the third item, fix $v \in [n]$ and $c \in [k]$. The number of hyperedges containing v together with another vertex of color c is $\sum_{u\in[n]\setminus\{v\}} \mathrm{codeg}_{\mathcal{H}}(u,v)\mathbf{1}_{\{\rho(u)=c\}}$. Since $\sum_{u\in[n]\setminus\{v\}} \mathrm{codeg}_{\mathcal{H}}(u,v) =$

$(t-1)d$ and every codegree is at most χ, we have:

$$\mathbb{E}\left[\sum_{u\in[n]\setminus\{v\}} \mathrm{codeg}_{\mathcal{H}}(u,v)\mathbf{1}_{\{\rho(u)=c\}}\right] = \frac{(t-1)d}{k},$$
$$\sum_{u\in[n]\setminus\{v\}} \mathrm{Var}\left(\mathrm{codeg}_{\mathcal{H}}(u,v)\mathbf{1}_{\{\rho(u)=c\}}\right) \leqslant \frac{1}{k}\sum_{u\in[n]\setminus\{v\}} \mathrm{codeg}_{\mathcal{H}}(u,v)^2 \leqslant \frac{(t-1)\chi d}{k}.$$

Thus by Proposition 3.3, except with probability 4^{-r} the number of hyperedges containing v together with another vertex of color c is at most $\frac{(t-1)d}{k} + 4C\left(\sqrt{\frac{r(t-1)\chi d}{k}} + r\chi\right) = O_t\left(\frac{d}{k} + \sqrt{\frac{\chi d\log n}{k}} + \chi\log n\right) = o(d)$. Taking a union bound over $v \in [n]$ and $c \in [k]$, the probability that the third item fails is at most $nk4^{-r} \leqslant \frac{1}{100}$.

For the fourth item, fix distinct $a, b \in [k]$. Partition the hyperedges of $\mathcal{H}$ into at most td matchings (this can be done as the line graph of $\mathcal{H}$ has degre at most $td-1$ and thus can be colored using at most td colors). Now fix one such matching $\mathcal{M}$. For every $e \in \mathcal{M}$, let Y_e be the indicator random variable that e is rainbow and uses colors a, b. We have $\mathbb{E}[Y_e] \leqslant t(t-1)/k^2$. Moreover, since the hyperedges in $\mathcal{M}$ are vertex disjoint, any r hyperedges of them involve tr distinct vertices. As ρ is tr-wise independent, every r indicator variables Y_e are jointly independent.

This immediately implies an upper bound on $Y_{\mathcal{M}}$ which is the sum of all Y_e for all $e \in \mathcal{M}$. By Proposition 3.3, except with probability 4^{-r}, we have:

$$Y_{\mathcal{M}} \leqslant \mu_{\mathcal{M}} + 4C\left(\sqrt{r\sum_{e\in\mathcal{M}}\mathrm{Var}(Y_e)} + r\right) \leqslant \mu_{\mathcal{M}} + 4C(\sqrt{r\mu_{\mathcal{M}}} + r) = O(\mu_{\mathcal{M}} + r),$$

where $\mu_{\mathcal{M}} := \mathbb{E}[Y_{\mathcal{M}}] = \sum_{e\in\mathcal{M}}\mathbb{E}[Y_e]$, and in the second inequality we used that $\mathrm{Var}(Y_e) = \mathbb{E}[Y_e](1-\mathbb{E}[Y_e]) \leqslant \mathbb{E}[Y_e]$ since $Y_e \in \{0,1\}$ and in the last equality we used $\sqrt{r\mu_{\mathcal{M}}} \leqslant r + \mu_{\mathcal{M}}$. Then, taking a union bound over all pairs $a, b \in [k]$ and all the matchings, we conclude that with probability at least $1 - 4^{-r}tdk^2 \geqslant 1 - 1/\mathrm{poly}(n) \geqslant 99/100$, this holds simultaneously for every pair of colors and every matching.

Summing over the td matchings and noting that the sum of $\mu_{\mathcal{M}}$ over all the matchings is at most $\frac{t(t-1)}{k^2}\cdot\frac{nd}{t} = O_t(nd/k^2)$, we get that for every pair of distinct colors $a, b \in [k]$, the number of rainbow hyperedges using both colors is at most $O\left(\sum_{\mathcal{M}}\mu_{\mathcal{M}} + tdr\right) = O_t(nd/k^2 + d\log n)$, where we used $r = O_t(\log n)$. We conclude that the probability the fourth item fails is at most $1/100$.

For the fifth item, fix numbers $\{q_v\}_{v\in[n]}$ as in the statement. For a fixed color $c \in [k]$, let X_v indicate that $\rho(v) = c$. The variables q_vX_v are r-wise independent, take values in $[0,d]$, and

$$\mathbb{E}\left[\sum_{v\in[n]} q_vX_v\right] = \frac{1}{k}\sum_{v\in[n]} q_v = \frac{nd}{tk}, \qquad \sum_{v\in[n]}\mathrm{Var}(q_vX_v) \leqslant \frac{1}{k}\sum_{v\in[n]} q_v^2 \leqslant \frac{nd^2}{tk}.$$

Thus, by Proposition 3.3 , except with probability 4^{-r},

$$\left|\sum_{v:\rho(v)=c} q_v - \frac{nd}{tk}\right| \leqslant 4Cd\left(\sqrt{\frac{rn}{tk}} + r\right) = o\left(\frac{nd}{tk}\right).$$

Taking a union bound over the colors, the probability that the fifth item fails is at most $k4^{-r} \leqslant \frac{1}{100}$.

Overall, by the union bound the probability that the chosen ρ fails to satisfy one of the items is at most $\frac{6}{100} < 1$, hence there is $\rho \in \mathcal{C}_{n,k}$ as required. □

3.2 The k-means Instance and Its Completeness

Fix a t-uniform d-regular hypergraph $\mathcal{H} = (V, E)$ on N vertices as in Theorem 3.2, and fix the family of colorings $\mathcal{C}_{N,k}$ from Theorem 3.4. We drop from $\mathcal{C}_{N,k}$ any coloring failing one of the first four items (this can be done as each one of them can be checked in polynomial time), and by Theorem 3.4 we still have $\mathcal{C}_{N,k} \neq \emptyset$.

For each coloring $\rho \in \mathcal{C}_{N,k}$, we construct a hypergraph $\mathcal{H}_\rho = (V, E_\rho)$ by deleting from $\mathcal{H}$ all hyperedges $e \in E$ in which two vertices are colored by the same color under ρ; we refer to such hyperedges as bad. Then by the second item in Theorem 3.4 we have that

$$|E_\rho| \geqslant M - 100\binom{t}{2}\frac{M}{k}. \tag{19}$$

Let s and R be integers that we will fix later. For each $\rho \in \mathcal{C}_{N,k}$, we construct a point set $P_\rho \subseteq \{0, 1, \sqrt{s}\}^{N+k}$, by including in it the point

$$p_e := \sqrt{s}\sum_{v \in e} \mathbf{e}_v + \sum_{c \in \rho(e)} \mathbf{e}_{N+c}$$

for each $e \in E_\rho$. In addition, we also include R/k copies of the point $\mathbf{e}_{N+c}$ for each $c \in [k]$. Note that each p_e has support of size $2t$ and $|P_\rho| = |E_\rho| + R = (1 + o(1))M$. In order to obtain the completeness and soundness claims of the theorem statement, we set $s = \sqrt{\frac{N^2}{k^3} + \frac{N\lceil \log_2 N \rceil}{k}}$ and $R = k\lceil d\sqrt{Ns/k} \rceil$.

This completes the description of the reduction, and to analyze it we need the following well known fact giving us a formula for the cost of a single cluster.

Fact 3.5. *For a nonempty finite multiset $A \subseteq \mathbb{R}^d$, the center minimizing its 1-means cost is $c = |A|^{-1}\sum_{x \in A} x$, and*

$$\min_{c \in \mathbb{R}^d} \sum_{x \in A} \|x - c\|_2^2 = \frac{1}{2|A|}\sum_{x,y \in A} \|x - y\|_2^2.$$

The following claim gives the completeness of the reduction.

Claim 3.6. *Suppose that there is $S \subseteq V$ of size N/t that touches each hyperedge of $\mathcal{H}$ exactly once. Then for some $\rho \in \mathcal{C}_{N,k}$ we have that $\mathrm{OPT}_k^{2,2}(P_\rho) \leqslant (ts + t - 1)(|P_\rho| - R) - (1 - o(1))skd$.*

Proof. For every hyperedge e we denote by $v(e)$ the unique vertex in $e \cap S$, and say that e is assigned to $v(e)$. Note that since $\mathcal{H}$ is d-regular, each vertex in S gets assigned exactly d hyperedges (before the bad hyperedges are deleted). We would like to cluster points e according to the color of $v(e)$, but we first need to ensure the clusters are roughly balanced in size. Towards this end define $q_v = d$ for $v \in S$ and $q_v = 0$ otherwise, so that q_v is precisely the number of hyperedges initially

assigned to v and $\sum_{v\in V} q_v = M$. Choose $\rho \in \mathcal{C}_{N,k}$ that also satisfies the last conclusion of Theorem 3.4 for these weights, and define the clusters $\mathcal{F}_c$ to consist of the points p_e with $\rho(v(e)) = c$, together with all R/k copies of $\mathbf{e}_{N+c}$, for $c \in [k]$. Letting q'_v be the number of not-bad hyperedges assigned to v, we have that $q'_v \leqslant q_v$ for every v so that, using $R = o(M)$, for all $c \in [k]$

$$|\mathcal{F}_c| = R/k + \sum_{v:\rho(v)=c} q'_v \leqslant R/k + \sum_{v:\rho(v)=c} q_v \leqslant (1+o(1))\frac{M}{k}. \tag{20}$$

Applying Fact 3.5, we get that

$$\mathrm{cost}(\mathcal{F}_c) = \sum_{p\in\mathcal{F}_c} \left\| p - \frac{1}{|\mathcal{F}_c|}\sum_{p'\in\mathcal{F}_c} p' \right\|_2^2 = (ts+t)(|\mathcal{F}_c| - R/k) + R/k - |\mathcal{F}_c|^{-1}\|\sum_{p\in\mathcal{F}_c} p\|_2^2, \tag{21}$$

and we now calculate $\|\sum_{p\in\mathcal{F}_c} p\|_2^2$.

The contribution of vertex coordinates: denoting $d_{\mathcal{F}_c,v} = |\{e \in E_\rho \mid p_e \in \mathcal{F}_c,\ v \in e\}|$, the contribution of vertex coordinates to $\|\sum_{p\in\mathcal{F}_c} p\|_2^2$ is $s\sum_{v\in V} d^2_{\mathcal{F}_c,v} \geqslant s\sum_{v:\rho(v)=c}(q'_v)^2$, where we used the fact that $d_{\mathcal{F}_c,v} \geqslant q'_v$ for v such that $\rho(v) = c$.

The contribution of color coordinates: denoting $r_{\mathcal{F}_c,a} = |\{p \in \mathcal{F}_c \mid p_{N+a} = 1\}|$, the contribution of color coordinates to $\|\sum_{p\in\mathcal{F}_c} p\|_2^2$ is $\sum_{a\in[k]} r^2_{\mathcal{F}_c,a} \geqslant r^2_{\mathcal{F}_c,c} = |\mathcal{F}_c|^2$.

Combining the bounds: plugging both bounds to (21) gives

$$\begin{aligned}\mathrm{cost}(\mathcal{F}_c) &\leqslant (ts+t-1)(|\mathcal{F}_c| - R/k) - \frac{s}{|\mathcal{F}_c|}\sum_{v:\rho(v)=c}(q'_v)^2 \\ &\leqslant (ts+t-1)(|\mathcal{F}_c| - R/k) - (1-o(1))\frac{sk}{M}\sum_{v:\rho(v)=c}(q'_v)^2,\end{aligned}$$

where we used (20). Summing over c gives

$$\mathrm{OPT}_k^{2,2}(P_\rho) \leqslant (ts+t-1)(|P_\rho| - R) - (1-o(1))\frac{sk}{M}\sum_v (q'_v)^2.$$

Note that $\sum_{v\in V} q_v^2 = d^2|S| = \frac{N}{t}d^2$. As $q_v \leqslant d$ for all v, removing a single edge from $\mathcal{H}$ drops this sum by at most $2d-1$, and iterating gives $\sum_{v\in V}(q'_v)^2 \geqslant \frac{N}{t}d^2 - 100\binom{t}{2}\frac{M}{k}\cdot(2d-1)$. Plugging this above and using (20) gives

$$\begin{aligned}\mathrm{OPT}_k^{2,2}(P_\rho) &\leqslant (ts+t-1)(|P_\rho| - R) - (1-o(1))skd + 200(1+o(1))\binom{t}{2}sd \\ &\leqslant (ts+t-1)(|P_\rho| - R) - (1-o(1))skd.\end{aligned}$$

□

3.3 Soundness Analysis

We now analyze the soundness of the construction from Section 3.2, and assume henceforth that $\mathcal{H}$ is as in the NO case. Fixing $\rho \in \mathcal{C}_{N,k}$, we show that if $\mathrm{OPT}_k^{2,2}(P_\rho) < (ts+t-1)(|P_\rho| - R) - (1 - 1/(8t))skd$, then one could find a set $S \subseteq V$ of size N/t covering more than $\frac{3}{4}M$ of the hyperedges of $\mathcal{H}$. The goal is to show that any low-cost clustering of P_ρ must basically look like the clustering we constructed in the completeness case, and we establish it via the following steps:

1. **Dominant color in each cluster:** we first establish that in a low-cost clustering of P_ρ, all but $o(M)$ points contain the dominant color of their cluster.
2. **Assigning edges to vertices:** the dominant color in each clusters identifies a distinguished vertex $v(e)$ for (almost all) edges $e \in E$, essentially reversing the analogous association from the completeness analysis. We argue that the collection of all distinguished elements is not too spread out, in the sense that many hyperedges e in a cluster pick that same vertex v.
3. **Choosing the set S:** finally, we pick our set S to be the collection of N/t vertices that were assigned to the most number of hyperedges, and we show that the previous item implies that S touches many of the hyperedges of $\mathcal{H}$.

We now move on to the formal proof. Fix a clustering $\mathcal{F}_1, \ldots, \mathcal{F}_k$ of P_ρ of cost smaller than $(ts+t-1)(|P_\rho| - R) - \left(1 - \frac{1}{8t}\right) skd$, and assume without loss of generality they are all nonempty and assume that if there are multiple copies of the same point, they all belong to the same cluster. For every $i \in [k]$, let m_i' be the number of points corresponding to hyperedges in $\mathcal{F}_i$, and define

$$m_i := |\mathcal{F}_i|, \qquad d_{i,v} := |\{e \in E_\rho \mid p_e \in \mathcal{F}_i,\ v \in e\}|, \qquad D_i := \frac{1}{m_i} \sum_{v \in V} d_{i,v}^2.$$

Let γ_i be the fraction of unordered pairs $\{p, p'\} \subseteq \mathcal{F}_i$ such that $\rho(p) \cap \rho(p') = \emptyset$, and let

$$\kappa_i := \frac{1}{\binom{m_i}{2}} \sum_{\{p,p'\} \subseteq \mathcal{F}_i} \max(|\rho(p) \cap \rho(p')| - 1, 0).$$

The following claim expresses the cost of each cluster in terms of the parameters above.

Claim 3.7. *We have* $\mathrm{cost}(\mathcal{F}_i) = (ts+t-1)m_i' - (t-1)\frac{m_i'}{m_i} - z_i$ *where* $z_i := sD_i - (m_i - 1)\gamma_i + (m_i - 1)\kappa_i$.

Proof. Writing $s_i = \sum_{p \in \mathcal{F}_i} p$, Fact 3.5 gives us that

$$\mathrm{cost}(\mathcal{F}_i) = \sum_{p \in \mathcal{F}_i} \|p\|_2^2 - \frac{1}{m_i}\|s_i\|_2^2 = m_i + (ts+t-1)m_i' - \frac{1}{m_i}\|s_i\|_2^2,$$

and the goal is to calculate $\|s_i\|_2^2$. Using the notation $d_{i,v}$ above and letting $r_{i,c}$ be the number of points of $\mathcal{F}_i$ containing color c, we have that $\|s_i\|_2^2 = \sum_{c \in [k]} r_{i,c}^2 + s \sum_{v \in V} d_{i,v}^2$, and we now calculate each

one of these sums. For the first one we have

$$\sum_{c\in[k]} r_{i,c}^2 = \sum_{c\in[k]} r_{i,c} + 2\sum_{c\in[k]} \binom{r_{i,c}}{2} = m_i + (t-1)m_i' + 2\sum_{\{p,p'\}\subseteq\mathcal{F}_i} |\rho(p)\cap\rho(p')|$$
$$= m_i + (t-1)m_i' + 2\binom{m_i}{2}(1-\gamma_i+\kappa_i).$$

and therefore $\sum_{c\in[k]} r_{i,c}^2 = m_i + (t-1)m_i' + m_i(m_i-1)(1-\gamma_i+\kappa_i)$. For the second one we have $\sum_{v\in V} d_{i,v}^2 = m_i D_i$, and combining and plugging above gives

$$\mathrm{cost}(\mathcal{F}_i) = m_i + (ts+t-1)m_i' - 1 - (t-1)\frac{m_i'}{m_i} - (m_i-1)(1-\gamma_i+\kappa_i) - sD_i,$$

and rearranging gives the claim. □

Summing Claim 3.7, we get the following result.

Claim 3.8. $\sum_{i=1}^k z_i \geqslant \left(1-\frac{1}{8t}-o(1)\right)skd$.

Proof. Summing Claim 3.7 over $i = 1,\ldots,k$

$$\sum_{i=1}^k z_i = \sum_{i=1}^k \left((ts+t-1)m_i' - (t-1)\frac{m_i'}{m_i} - \mathrm{cost}(\mathcal{F}_i)\right)$$
$$\geqslant (ts+t-1)(|P_\rho|-R) - (t-1)k - \left((ts+t-1)(|P_\rho|-R) - \left(1-\frac{1}{8t}\right)skd\right).$$

Using $(t-1)k = o(skd)$ and simplifying gives the result. □

3.3.1 Dominant Color in Each Cluster

Our next goal is to show that the properties of the coloring ρ imply that κ_i is negligible. In that case, Claim 3.8 implies that on average D_i is large, giving that $d_{i,v}$ is abnormally large for some v's.

Claim 3.9. *For every $i\in[k]$ it holds that $(m_i-1)\kappa_i = o(sd)$.*

Proof. For distinct colors a,b, let $q_{a,b}$ denote the number of points in $\mathcal{F}_i$ containing both a and b. By the properties of ρ and the choice of s, we have $q_{a,b} = O_t(Nd/k^2 + d\log N) = o(sd)$, since each of the additionally introduced point has only one color. It follows that

$$\sum_{\{p,p'\}\subseteq\mathcal{F}_i} \max(|\rho(p)\cap\rho(p')|-1,0) \leqslant \sum_{a<b}\binom{q_{a,b}}{2} \leqslant o(sd)\cdot\sum_{a<b} q_{a,b} \leqslant o\left(sd\binom{t}{2}m_i\right).$$

Dividing by $m_i/2$ gives $(m_i-1)\kappa_i \leqslant o\left(\binom{t}{2}sd\right) = o(sd)$. □

For every $i\in[k]$, let c_i be a color appearing in the maximum number of points of $\mathcal{F}_i$, and let

$$h_i = |\{p\in\mathcal{F}_i : c_i\in\rho(p)\}|, \qquad R_i = m_i - h_i.$$

Thus, R_i is the number of points in $\mathcal{F}_i$ that do not contain its dominant color. We next show that in total, only few points do not contain the dominant color of their cluster.

Claim 3.10. *We have* $\sum_{i=1}^{k} R_i = O_t(skd) = o(M)$.

Proof. Our main goal is to show that

$$(m_i - 1)\gamma_i \geqslant \frac{R_i}{4t} - o(sd), \tag{22}$$

and we first explain why that completes the proof. Rearranging and summing gives

$$\begin{aligned}\sum_{i=1}^{k} R_i &\leqslant o(skd) + 4t\sum_{i=1}^{k}(m_i-1)\gamma_i \\ &\leqslant o(skd) + 4t\sum_{i=1}^{k}(sD_i - z_i + (m_i-1)\kappa_i) \leqslant o(skd) + 4t\sum_{i=1}^{k}(sD_i - z_i),\end{aligned}$$

where we used Claim 3.9. Note that $\sum_{i=1}^{k} D_i \leqslant \sum_{i=1}^{k} \frac{d}{|\mathcal{F}_i|}\sum_{v\in V} d_{i,v} \leqslant tdk$, and by Claim 3.8 we have $\sum_{i=1}^{k} z_i \geqslant 0$, so the right hand side is at most $o(skd) + O_t(skd) = O_t(skd) = o(M)$, since $sk = o(N)$.

We now establish (22), and for that we denote by $Z_i = \binom{m_i}{2}\gamma_i$ the number of unordered pairs of points in $\mathcal{F}_i$ having disjoint color sets. Fix $p \in \mathcal{F}_i$ not containing c_i, and note that for each of its at most t colors, at most $o(sd)$ points contain both that color and c_i. Thus p shares a color with at most $o(sd)$ of the h_i points containing c_i, and hence

$$Z_i \geqslant R_i(h_i - o(sd)).$$

If $h_i \geqslant \frac{m_i}{2t}$ then we immediately get (22), so we henceforth assume that $h_i < \frac{m_i}{2t}$. Since c_i is a most frequent color, we have $r_{i,c} \leqslant h_i$ for every c, so the number of pairs of points whose color sets intersect is at most $\sum_{c\in[k]}\binom{r_{i,c}}{2} \leqslant \frac{h_i}{2}\sum_{c\in[k]} r_{i,c} \leqslant \frac{th_im_i}{2}$. Thus,

$$Z_i \geqslant \binom{m_i}{2} - \frac{th_im_i}{2} \geqslant \binom{m_i}{2} - \frac{m_i^2}{4} = \frac{m_i^2 - 2m_i}{4} \geqslant m_i\frac{R_i}{8t},$$

and multiplying by $\frac{2}{m_i}$ implies (22). □

3.3.2 Assigning Hyperedges to Vertices

For every hyperedge e with $p_e \in \mathcal{F}_i$ containing c_i, let $v(e)$ be the unique vertex in e such that $\rho(v(e)) = c_i$. For every vertex v with $\rho(v) = c_i$, let $a_{i,v} := |\{e : p_e \in \mathcal{F}_i,\ v(e) = v\}|$. We wish to show that these assignments are concentrated on relatively few vertices. First, note that in the completeness case, a cluster contains hyperedges assigned to vertices of a fixed color, leading to a saving of roughly sd (due to the coordinates corresponding to these vertices being large in $\sum_{p\in\mathcal{F}_i} p$), and we want to argue that something similar must happen in our case. Claim 3.8 gives a lower bound on the saving in the clusters, but in principle it could be the case that all of the savings come from a single cluster. The following claim shows that the saving z_i exceeds sd by much in only $o(k)$ clusters, thereby eliminating this possibility and asserting that the saving must be spread among the different clusters.

Claim 3.11. *For every $i \in [k]$ we have $z_i \leqslant (t + o(1))sd$, and for all but $o(k)$ of them we have $z_i \leqslant (1 + o(1))sd$.*

Proof. The first bound follows from $D_i \leqslant td$ and Claim 3.9. To prove the second bound, we need to upper bound D_i, and to do that we first consider an auxiliary parameter D'_i. Consider only the hyperedges corresponding to points in $\mathcal{F}_i$ that contain the dominant color c_i, let $d'_{i,v}$ denote the degree of v among these hyperedges, and define $D'_i := \frac{1}{m_i}\sum_{v \in V}(d'_{i,v})^2$. Every such hyperedge contains exactly one vertex of color c_i. Since $d'_{i,v} \leqslant d$ for every v and $\sum_{v:\rho(v)=c_i} d'_{i,v} \leqslant h_i$, the contribution of vertices of color c_i is at most $\sum_{v:\rho(v)=c_i}(d'_{i,v})^2 \leqslant dh_i$. Moreover, by the third item of Theorem 3.4, every vertex of color different from c_i has degree at most $o(d)$ among these hyperedges. Since there are at most $(t-1)h_i$ incidences with such vertices, their contribution is at most $o(d)h_i$. Hence

$$D'_i \leqslant \frac{dh_i + o(d)h_i}{m_i} \leqslant d + o(d).$$

We now add back the remaining at most R_i hyperedges. Adding one hyperedge increases the degree of each of its t vertices by one, and hence increases the sum of squared degrees by at most $2td$. Therefore

$$D_i \leqslant D'_i + \frac{2tdR_i}{m_i} \leqslant d + o(d) + \frac{2tdR_i}{m_i}.$$

If $m_i \geqslant 8t^2sd$, then we get that

$$z_i \leqslant sD_i - (m_i - 1)\gamma_i + (m_i - 1)\kappa_i \leqslant sd + o(sd) + \frac{R_i}{4t} - (m_i - 1)\gamma_i + o(sd) \leqslant (1 + o(1))sd,$$

where we used (22) to bound $\frac{R_i}{4t} - (m_i - 1)\gamma_i$ and then Claim 3.9. It remains to show that only $o(k)$ clusters have fewer than $8t^2sd$ points. For each color c, consider the cluster containing all R/k copies of $\mathbf{e}_{N+c}$. If its dominant color is not c, these R/k points are counted in R_i. By Claim 3.10, this happens for at most $O_t(sk^2d/R) = o(k)$ colors. The other $k - o(k)$ colors give distinct clusters, each containing at least $R/k = \omega(sd)$ points, as required. □

We already know that the total saving $\sum_{i=1}^{k} z_i$ is large by Claim 3.8, and the next claim shows that it is still large when we weight the clusters according to the popularity of the dominant color it them.

Claim 3.12. *We have $\sum_{i=1}^{k} h_i z_i \geqslant \left(\frac{7}{8} - o(1)\right) sMd$.*

Proof. By the balance of the color classes and $R = o(M)$, we have $h_i \leqslant d|\rho^{-1}(c_i)| + R/k \leqslant (t + o(1))\frac{M}{k}$, while Claim 3.11 gives $z_i \leqslant (1 + o(1))sd$ for all but $o(k)$ clusters, and $z_i \leqslant (t + o(1))sd$ for the remaining clusters. It follows that

$$\sum_{i=1}^{k}((1 + o(1))sd - z_i)h_i \leqslant (t + o(1))\frac{M}{k}\sum_{i=1}^{k}((1 + o(1))sd - z_i) + o(sMd) \leqslant \left(\frac{1}{8} + o(1)\right) sdM,$$

where the $o(sMd)$ term accounts for the $o(k)$ remaining clusters, and we used Claim 3.8 in the last transition. Rearranging gives $\sum_{i=1}^{k} h_i z_i \geqslant (1 + o(1))sd \sum_{i=1}^{k} h_i - \left(\frac{1}{8} + o(1)\right) sdM$. Using Claim 3.10 to note that $\sum_i h_i = |P_\rho| - \sum_i R_i = (1 + o(1))M$ finishes the proof. □

We now conclude that dominant vertices are not too spread.

Claim 3.13. *We have* $\sum_i \sum_{v:\rho(v)=c_i} a_{i,v}^2 \geqslant \left(\frac{7}{8} - o(1)\right) Md$.

Proof. Consider only the hyperedges corresponding to points of $\mathcal{F}_i$ that have the color c_i, and let $d'_{i,v}$ be the degree of v among them. Since removing one hyperedge decreases the sum of squared degrees by at most $2td$, we have

$$\sum_{v \in V} (d'_{i,v})^2 \geqslant m_i D_i - 2tdR_i.$$

For vertices of color c_i, we have $d'_{i,v} = a_{i,v}$. On the other hand, as in the proof of Claim 3.11, vertices of other colors contribute at most $o(d)h_i$ to the sum of squared degrees. Consequently,

$$\sum_{v:\rho(v)=c_i} a_{i,v}^2 \geqslant m_i D_i - 2tdR_i - o(d)h_i.$$

By the definition of z_i and $\gamma_i \geqslant 0$, we have $D_i \geqslant (z_i - (m_i - 1)\kappa_i)/s$, and since $m_i \geqslant h_i$ we get that

$$\sum_{v:\rho(v)=c_i} a_{i,v}^2 \geqslant \frac{h_i z_i}{s} - \frac{h_i(m_i - 1)\kappa_i}{s} - 2tdR_i - o(d)h_i.$$

Summing over i, the first term is at least $(7/8 - o(1))Md$ by Claim 3.12, while the remaining terms are $o(Md)$ by Claim 3.9, Claim 3.10, and $\sum_i h_i = O(M)$. □

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3.3.3 Choosing the Set S

For every $v \in V$, let $A_v = \sum_{i:c_i=\rho(v)} a_{i,v}$ be the total number of hyperedges assigned to v. Since all $a_{i,v}$ are nonnegative, we get from Claim 3.13 that

$$\sum_{v \in V} A_v^2 \geqslant \sum_{v \in V} \sum_{i:c_i=\rho(v)} a_{i,v}^2 \geqslant \left(\frac{7}{8} - o(1)\right) Md. \tag{23}$$

Let S consist of the N/t vertices with the largest values of A_v. The following claim completes the soundness analysis.

Claim 3.14. *S covers more than $3M/4$ hyperedges.*

Proof. For every $v \in V$, define $x_v := A_v/d$. Since $A_v \leqslant d$ and $\sum_v A_v \leqslant M = Nd/t$, we have

$$0 \leqslant x_v \leqslant 1, \qquad \sum_v x_v \leqslant \frac{N}{t}.$$

Thus $(x_v)_{v \in V}$ is a fractional choice of at most N/t vertices. Since S consists of the N/t vertices with the largest values of A_v, it captures at least as much load as this fractional choice, so using (23) we get

$$\sum_{v \in S} A_v \geqslant \sum_v x_v A_v = \frac{1}{d} \sum_v A_v^2 \geqslant \left(\frac{7}{8} - o(1)\right) M > \frac{3M}{4}.$$ □

4 Lower Bound for Euclidean k-median under ETH

In this section we improve on the range of parameters of [KM26] for Euclidean k-median and prove the following.

Theorem 4.1. *Assuming ETH, there is a constant $C > 0$ such that the following holds. Let $\widetilde{\varepsilon}\colon \mathbb{N} \to (0,1)$, $\widetilde{k}\colon \mathbb{N} \to \mathbb{N}$ be polynomial time computable functions of the input size satisfying:*

1. *$(\log n)^2 \leqslant \frac{\widetilde{k}(n)}{\widetilde{\varepsilon}(n)} = o\left(\frac{n}{\log^L n}\right)$ and the sequence $\frac{\widetilde{k}(n)}{\widetilde{\varepsilon}(n)}$ has bounded consecutive ratios.*
2. *$\widetilde{k}(n) = \omega(1)$ and $\widetilde{\varepsilon}(n) = o(1/\log n)$.*

Then for sufficiently large n, setting $\varepsilon = \widetilde{\varepsilon}(n)$ and $k = \widetilde{k}(n)$, Euclidean k-median on n points in $\mathbb{R}^d$ has no $(1+\varepsilon)$-approximation running in time $2^{(k/\varepsilon)/(\log(k/\varepsilon))^C}\operatorname{poly}(n,d)$.

In the proof of Theorem 4.1 we will transform the k-means lower bound from Theorem 3.1. The main idea is to give each point a new, unique coordinate of a large magnitude which we denote by T (a similar idea was used in [She21]). For a center that uses none of these new coordinates, a point whose original squared distance to the center is x now has distance

$$\sqrt{T^2 + x} = T\sqrt{1 + \frac{x}{T^2}} = T\left(1 + \frac{x}{2T^2} + O\left(\frac{x}{T^2}\right)^2\right) = T + \frac{1}{2T}x + O\left(\frac{x^2}{T^3}\right)$$

to its center. Thus, if x is significantly smaller than T^2, we get that the distance of the point from the center relates linearly to x. This suggest that as long as the centers do not use the new coordinates, one can directly relate the k-median value of the new instance with the k-mean value of the original instance. This turns out to be true but requires a much more delicate analysis due to the accumulation of errors in the Taylor expansion above across all of points. To finish the proof, one has to argue that the centers cannot benefit much from using the private coordinates, which holds because each point gets its own private coordinate.

Construction. Fix a t-uniform d-regular hypergraph $\mathcal{H} = (V, E)$ on N vertices as in Theorem 3.2, and fix the family of colorings $\mathcal{C}_{N,k}$ from Theorem 3.4. We drop from $\mathcal{C}_{N,k}$ any coloring failing one of the first four items (this can be done as each one of them can be checked in in polynomial time), and by Theorem 3.4 we still have $\mathcal{C}_{N,k} \neq \emptyset$. For each retained coloring $\rho \in \mathcal{C}_{N,k}$, let $p_1, \ldots, p_m$ be the points from P_ρ from Section 3.2, counting repeated points separately, and use the integers s, R from that construction. We first give each color point the same squared norm as a hyperedge point. For each of the R color points, append t coordinates equal to $\sqrt{s}$ and $t-1$ coordinates equal to one, using different coordinates for different point occurrences and setting all the other new coordinates to zero. We continue to denote the resulting points by $p_1, \ldots, p_m$, while P_ρ denotes the original multiset. Each p_j now has squared norm $ts + t$. Its color set $\rho(p_j)$ is unchanged.

For an integer $T > 0$ (to be determined later), define the new point set $Q_\rho := \{q_1, \ldots, q_m\} \subseteq \mathbb{R}^{N+k+(2t-1)R+m}$ as follows:

$$q_j := (p_j, T\mathbf{e}_j) \in \mathbb{R}^{N+k+(2t-1)R+m} \quad \forall j \in [m].$$

Thus each point has $2t+1$ nonzero coordinates and we refer to the last m coordinates as private coordinates. This completes the description of the reduction, and the heart of the analysis of the reduction is in the following lemma.

Lemma 4.2. *For every $t \geqslant 2$ and $\eta > 0$, we can choose T to be an integer multiple of $\sqrt{s}$, with $T/\sqrt{s}$ depending only on t and η, such that for every coloring $\rho \in \mathcal{C}_{N,k}$ and sufficiently large N, we have:*

$$\left|\mathrm{OPT}_k^{(2,1)}(Q_\rho) - \frac{\mathrm{OPT}_k^{(2,2)}(P_\rho) + (ts+t-1)R}{2r} - \frac{2r^2-(ts+t-1)}{2r}|P_\rho|\right| \leqslant \frac{skd}{2r}\eta,$$

where $r = \sqrt{T^2+ts+t-1}$.

4.1 Proof of Theorem 4.1 Assuming Lemma 4.2

Suppose towards contradiction that such an approximation algorithm exists. Take a t-uniform hypergraph $\mathcal{H}$ from Theorem 2.2 in [KM26] with error parameter $\frac{1}{100}$, N vertices and degree d, and let $M = Nd/t$. As before, we take the first q such that $\frac{\widetilde{k}(q)}{\widetilde{\varepsilon}(q)} \geqslant BtN$, where B is a sufficiently large constant to be determined. Using the same argument as in Theorem 3.1 we have that $q \geqslant N$, $\frac{\widetilde{k}(q)}{\widetilde{\varepsilon}(q)} = \Theta(N)$ and $M = o(q)$, and we denote for notational simplicity $k = \widetilde{k}(q)$, $\varepsilon = \widetilde{\varepsilon}(q)$ and $k' = k-1$. We take the collection of colorings $\mathcal{C}_{N,k'}$, from Theorem 3.4, and we only keep colorings satisfying the first four items.

Choose $\eta = 1/(64t)$ and apply Lemma 4.2 to get T, $r = \sqrt{T^2+ts+t-1}$ for the collections $(Q_\rho)_{\rho\in\mathcal{C}_{N,k'}}$ above, where s, R are chosen with k' clusters. Recalling that $|Q_\rho| \leqslant (1+o(1))M = o(q)$, we pad each Q_ρ by setting $W = \lfloor (q-1)/|Q_\rho| \rfloor$ and forming Q'_ρ by including W copies of each point from Q_ρ where we append a new 0 coordinate, and including $q - W|Q_\rho|$ copies of the point $x = Hq\mathbf{e}_{\text{new}}$, where $H > 2\sqrt{T^2+ts+t}$ is an integer. As in the proof of Theorem 3.1 we get that

$$\mathrm{OPT}_k^{(2,1)}(Q'_\rho) = W\,\mathrm{OPT}_{k'}^{(2,1)}(Q_\rho),$$

and we now use the guarantee of Lemma 4.2 to calculate the gap between the YES and the NO instances. In the YES case, by Theorem 3.2, there is some ρ such that $\mathrm{OPT}_{k'}^{(2,2)}(P_\rho) \leqslant (ts+t-1)(|P_\rho|-R) - (1-o(1))sk'd$, so by Lemma 4.2

$$\begin{aligned}
\mathrm{OPT}_{k'}^{(2,1)}(Q_\rho) &\leqslant \left(r - \frac{ts+t-1}{2r}\right)|P_\rho| + \frac{\mathrm{OPT}_{k'}^{(2,2)}(P_\rho) + (ts+t-1)R}{2r} + \frac{\eta sk'd}{2r} \\
&\leqslant r|P_\rho| - (1-\eta-o(1))\,\frac{sk'd}{2r} \\
&\leqslant r|P_\rho| - \left(1-\frac{1}{32t}\right)\frac{sk'd}{2r},
\end{aligned}$$

which we denote by U_ρ. In the last transition we used the fact that N is sufficiently large. In the NO case, by Theorem 3.2, for every $\rho \in \mathcal{C}_{N,k'}$ we have $\mathrm{OPT}_{k'}^{(2,2)}(P_\rho) \geqslant (ts+t-1)(|P_\rho|-R) -$

$\left(1-\frac{1}{8t}\right) sk'd$, so by Lemma 4.2

$$\begin{aligned}\mathrm{OPT}_{k'}^{(2,1)}(Q_\rho) &\geqslant \left(r-\frac{ts+t-1}{2r}\right)|P_\rho| + \frac{\mathrm{OPT}_{k'}^{(2,2)}(P_\rho)+(ts+t-1)R}{2r} - \frac{\eta sk'd}{2r}\\ &\geqslant r|P_\rho| - \left(1-\frac{1}{8t}+\eta\right)\frac{sk'd}{2r}\\ &= r|P_\rho| - \left(1-\frac{7}{64t}\right)\frac{sk'd}{2r},\end{aligned}$$

which we denote by L_ρ. It follows that for each ρ, the ratio between these two bounds after padding is also L_ρ/U_ρ, and we now calculate it. Note that $L_\rho - U_\rho = \frac{5sk'd}{128tr} > \frac{sk'd}{32tr}$, so

$$\frac{L_\rho}{U_\rho} \geqslant 1 + \frac{sk'd}{32trU_\rho} = 1+\Theta\left(\frac{BsN\varepsilon d}{r^2M}\right) = 1+\Theta\left(\frac{Bst}{r^2}\right)\varepsilon > 1+\varepsilon$$

for sufficiently large B, since $r^2/s = T^2/s + t + (t-1)/s$ is bounded by a constant depending only on t and η. Running the assumed algorithm on each Q'_ρ and comparing its returned cost with $(1+\varepsilon)WU_\rho$ distinguishes the two cases. The construction and running-time bounds are the same as in the proof of Theorem 3.1, contradicting ETH.

4.2 Proof of Lemma 4.2

Fix a coloring $\rho \in \mathcal{C}_{N,k}$ and denote $m = |P_\rho| = |Q_\rho|$. By Fact 3.5, the modification of the color points increases the 1-means cost of a cluster of m_i points by $(ts+t-1)(1-1/m_i)$ per color point. Thus the total increase for any partition into at most k clusters is between $(ts+t-1)(R-k)$ and $(ts+t-1)R$. We prove the upper bound and the lower bound separately.

4.2.1 Proof of the Upper bound

Consider an optimal k-means clustering of P_ρ, and let $s_1,\dots,s_k$ be the centroids of the corresponding clusters after modifying the color points. Let $s'_1,\dots,s'_k$ be the points $s_1,\dots,s_k$ where we append 0's on all of the private coordinates, and consider it as centers for Q_ρ. Since the second order derivatives of $x \mapsto \sqrt{T^2+x}$ are negative, we get by Taylor approximation around $ts+t-1$ that

$$\sqrt{T^2+x} \leqslant r + \frac{1}{2r}(x-(ts+t-1)).$$

Applying this inequality to each point in Q_ρ and then summing gives

$$\begin{aligned}\mathrm{OPT}_k^{(2,1)}(Q_\rho) &\leqslant rm + \frac{\mathrm{OPT}_k^{(2,2)}(P_\rho)+(ts+t-1)R-(ts+t-1)m}{2r}\\ &= \left(r-\frac{ts+t-1}{2r}\right)m + \frac{\mathrm{OPT}_k^{(2,2)}(P_\rho)+(ts+t-1)R}{2r},\end{aligned}$$

where we used the preceding bound on the change in the k-means cost.

4.2.2 Proof of the Lower bound

Fix an optimal k-median clustering of Q_ρ, and let $\mathcal{F}_1, \ldots, \mathcal{F}_k$ be the corresponding clusters of indices in $[m]$. We will use the following notations from Section 3.3: let $m_i = |\mathcal{F}_i|$, let c_i be the most popular color in $\mathcal{F}_i$, let R_i be the number of points in $\mathcal{F}_i$ that do not contain the color c_i and let $\mu_i = \frac{1}{m_i}\sum_{j\in\mathcal{F}_i} p_j$. We write $\mathrm{cost}(\mathcal{F}_i)$ for the 1-means cost of the modified points $\{p_j : j \in \mathcal{F}_i\}$, and use $D_i, \gamma_i, \kappa_i, z_i$ for the quantities from that analysis, computed before modifying the color points. The cluster-cost identity and the bounds on γ_i and κ_i apply to these clusters without requiring copies of the same point to belong to the same cluster. The following lemma breaks the cost of each cluster into its k-means cost, the effect of the private coordinates, and the terms arising from the Taylor approximation and the choice of center.

Lemma 4.3. *For every center (y, b), we have*

$$\sum_{j\in\mathcal{F}_i} \|q_j - (y,b)\|_2 = \underbrace{\left(r - \frac{ts+t-1}{2r}\right) m_i + \frac{\mathrm{cost}(\mathcal{F}_i)}{2r}}_{\text{main term}} + \underbrace{\frac{m_i}{2r}\|y-\mu_i\|_2^2}_{\text{distance from } \mu_i} - \underbrace{\frac{1}{2r}\sum_{j\in\mathcal{F}_i}\left(\sqrt{T^2+\|p_j-y\|_2^2} - r\right)^2}_{\text{Taylor error}} - \underbrace{\sum_{j\in\mathcal{F}_i}\left(\sqrt{T^2+\|p_j-y\|_2^2} - \|q_j-(y,b)\|_2\right)}_{\text{effect of private coordinates}}.$$

Proof. Our goal is to relate the k-median cost of the cluster to its k-means cost. We first consider a center $(y, 0)$ that does not use the private coordinates. In this case, the distance of q_j from the center is

$$\|q_j - (y,0)\|_2 = \sqrt{T^2 + \|p_j - y\|_2^2}.$$

Its first-order Taylor approximation around $ts + t - 1$ is $r + \frac{\|p_j-y\|_2^2 - (ts+t-1)}{2r}$, and we want to calculate the difference between them explicitly. Recalling that $r^2 = T^2 + ts + t - 1$, we have for every $x \geqslant 0$

$$r + \frac{x - (ts+t-1)}{2r} - \sqrt{T^2+x} = \frac{(\sqrt{T^2+x} - r)^2}{2r}.$$

Thus, summing over the cluster gives

$$\sum_{j\in\mathcal{F}_i} \|q_j - (y,0)\|_2 = \left(r - \frac{ts+t-1}{2r}\right) m_i + \frac{1}{2r}\sum_{j\in\mathcal{F}_i}\|p_j-y\|_2^2 - \frac{1}{2r}\sum_{j\in\mathcal{F}_i}\left(\sqrt{T^2+\|p_j-y\|_2^2} - r\right)^2.$$

The first sum of squared distances is almost the k-means cost of the cluster, except that y need not be its center. By Fact 3.5, we have

$$\sum_{j\in\mathcal{F}_i}\|p_j-y\|_2^2 = \mathrm{cost}(\mathcal{F}_i) + m_i\|y-\mu_i\|_2^2,$$

and plugging this above we get

$$\sum_{j\in\mathcal{F}_i}\|q_j-(y,0)\|_2=\left(r-\frac{ts+t-1}{2r}\right)m_i+\frac{1}{2r}\mathrm{cost}(\mathcal{F}_i)+\frac{m_i}{2r}\|y-\mu_i\|_2^2 \\ -\frac{1}{2r}\sum_{j\in\mathcal{F}_i}\left(\sqrt{T^2+\|p_j-y\|_2^2}-r\right)^2.$$

Finally, we account for the fact that the centers of $\mathcal{F}_i$ may use the private coordinates. For an arbitrary center (y,b), we have

$$\sum_{j\in\mathcal{F}_i}\|q_j-(y,b)\|_2=\sum_{j\in\mathcal{F}_i}\|q_j-(y,0)\|_2-\sum_{j\in\mathcal{F}_i}\left(\sqrt{T^2+\|p_j-y\|_2^2}-\|q_j-(y,b)\|_2\right).$$

Combining this with the previous calculation gives exactly the claimed decomposition. □

We now bound the error terms in Lemma 4.3. We start with the contribution of the private coordinates.

Claim 4.4. *For an optimal center (y,b) of $\mathcal{F}_i$, we have*

$$\sum_{j\in\mathcal{F}_i}\left(\sqrt{T^2+\|p_j-y\|_2^2}-\|q_j-(y,b)\|_2\right)\leqslant 2T.$$

Proof. As an optimal 1-median center can be chosen in the convex hull of the points in its cluster, we assume that (y,b) is in the convex hull of $\{q_j : j\in\mathcal{F}_i\}$, and the coordinates of b are non-negative and sum up to T. For every $j\in\mathcal{F}_i$, we have $\|q_j-(y,0)\|_2^2-\|q_j-(y,b)\|_2^2=2Tb_j-\|b\|_2^2\leqslant 2Tb_j$, and as $\|q_j-(y,0)\|_2\geqslant T$, we get

$$\|q_j-(y,0)\|_2-\|q_j-(y,b)\|_2\leqslant 2b_j.$$

Summing over the cluster gives the claim. □

We next bound the Taylor error term on Lemma 4.3, and we show that its negative contribution is almost entirely offset by the distance to μ_i.

Claim 4.5. *For sufficiently large $T/\sqrt{s}$, and every y in the convex hull of the points of $\mathcal{F}_i$, we have*

$$\frac{1}{2r}\sum_{j\in\mathcal{F}_i}\left(\sqrt{T^2+\|p_j-y\|_2^2}-r\right)^2\leqslant\frac{m_i}{2r}\|y-\mu_i\|_2^2+O_t\left(\frac{s(R_i+sd)}{T^3}\right).$$

Proof. Since $r^2=T^2+ts+t-1$, we have

$$\frac{1}{2r}\left(\sqrt{T^2+x}-r\right)^2=\frac{(x-(ts+t-1))^2}{2r(\sqrt{T^2+x}+r)^2}\leqslant\frac{(x-(ts+t-1))^2}{8T^3}. \tag{24}$$

Thinking of $x=\|p_j-y\|_2^2$, our goal is to show that $\|p_j-y\|_2^2$ are typically close to $ts+t-1$. Towards this end, let $u_i=e_{N+c_i}$ be the unit vector corresponding to the dominant color c_i, and

note that $\|p_j - u_i\|_2^2$ is $ts + t - 1$ if p_j contains c_i, and is $ts + t + 1$ otherwise. In particular, only R_i points have squared distance different from $ts + t - 1$. We get

$$\begin{aligned}\|p_j - y\|_2^2 - (ts + t - 1) &= \|p_j - u_i\|_2^2 - (ts + t - 1) - 2\langle p_j - u_i, y - u_i\rangle + \|y - u_i\|_2^2 \\ &= 2 \cdot 1_{c_i \notin \rho(p_j)} - 2\langle p_j - u_i, y - u_i\rangle + \|y - u_i\|_2^2.\end{aligned}$$

Using Cauchy-Schwarz and the fact that $\|p_j - u_i\|_2, \|y - u_i\|_2 \leqslant \sqrt{ts + t + 1}$ gives

$$|\|p_j - y\|_2^2 - (ts + t - 1)| \leqslant 2 \cdot 1_{c_i \notin \rho(p_j)} + 3\sqrt{ts + t + 1}\|y - u_i\|_2.$$

Squaring, using $(a + b)^2 \leqslant 2a^2 + 2b^2$ and summing gives

$$\sum_{j \in \mathcal{F}_i} \left(\|p_j - y\|_2^2 - (ts + t - 1)\right)^2 \leqslant 8R_i + 18(ts + t + 1)m_i\|y - u_i\|_2^2.$$

We next upper bound $\|y - u_i\|_2^2$. Using the triangle inequality we get

$$m_i\|y - u_i\|_2^2 \leqslant 2m_i\|y - \mu_i\|_2^2 + 2m_i\|\mu_i - u_i\|_2^2.$$

By Fact 3.5, Claim 3.7, and the change in cost from modifying the color points, we get

$$\begin{aligned}m_i\|\mu_i - u_i\|_2^2 &= \sum_{j \in \mathcal{F}_i} \|p_j - u_i\|_2^2 - \text{cost}(\mathcal{F}_i) \\ &= (ts + t - 1)m_i + 2R_i - \text{cost}(\mathcal{F}_i) \leqslant 2R_i + (ts + t - 1) + z_i,\end{aligned}$$

which is at most $2R_i + O_t(sd)$ using Claim 3.9 and $D_i \leqslant td$. Plugging this above and then propagating further gives

$$\sum_{j \in \mathcal{F}_i} \left(\|p_j - y\|_2^2 - (ts + t - 1)\right)^2 \leqslant O_t(sR_i) + 18(ts + t + 1)(2m_i\|y - \mu_i\|_2^2 + O_t(sd)).$$

Plugging $x = \|p_j - y\|_2^2$ in (24) and using the above to upper bound the right hand side of (24), we get that the left hand side of the claim is upper bounded by

$$O_t\left(\frac{s(R_i + sd)}{T^3}\right) + \frac{9(ts + t + 1)}{2T^3}m_i\|y - \mu_i\|_2^2 \leqslant O_t\left(\frac{s(R_i + sd)}{T^3}\right) + \frac{m_i}{2r}\|y - \mu_i\|_2^2,$$

where the last inequality is for large enough $T/\sqrt{s}$. □

Plugging Claim 4.4 and Claim 4.5 into Lemma 4.3 and summing over i gives the bound

$$\text{OPT}_k^{(2,1)}(Q_\rho) \geqslant \left(r - \frac{ts + t - 1}{2r}\right)\sum_{i=1}^k m_i + \frac{1}{2r}\sum_{i=1}^k \text{cost}(\mathcal{F}_i) - \frac{s}{T^3}O_t\left(sdk + \sum_{i=1}^k R_i\right) - 2Tk, \quad (25)$$

and we next bound $\sum_{i=1}^{k} R_i$. Rearranging (22) we get that

$$\begin{aligned}\sum_{i=1}^{k} R_i &\leqslant o(skd) + 4t\sum_{i=1}^{k}(m_i - 1)\gamma_i \\ &= o(skd) + 4t\sum_{i=1}^{k}(sD_i - z_i + (m_i - 1)\kappa_i) \leqslant O_t(skd) - 4t\sum_{i=1}^{k} z_i,\end{aligned}$$

where we used Claim 3.9 and $D_i \leqslant td$ in the last transition. Using $z_i \geqslant (ts + t - 1)m_i - (ts + t - 1) - \mathrm{cost}(\mathcal{F}_i)$, as above, and plugging that into (25) gives

$$\begin{aligned}\mathrm{OPT}_k^{(2,1)}(Q_\rho) &\geqslant rm + \left(\frac{1}{2r} - \frac{O_t(s)}{T^3}\right)\left(\sum_{i=1}^{k}\mathrm{cost}(\mathcal{F}_i) - (ts + t - 1)m\right) - O_t\left(\frac{s^2kd}{T^3}\right) - 2Tk \\ &\geqslant rm + \left(\frac{1}{2r} - \frac{O_t(s)}{T^3}\right)\left(\mathrm{OPT}_k^{(2,2)}(P_\rho) - (ts + t - 1)(m - R)\right) \\ &\quad - O_t\left(\frac{s^2kd}{T^3}\right) - 2Tk - \frac{(ts + t - 1)k}{2r},\end{aligned}$$

where we used $\frac{1}{2r} - \frac{O_t(s)}{T^3} \geqslant 0$ for sufficiently large $T/\sqrt{s}$, and $\sum_{i=1}^{k}\mathrm{cost}(\mathcal{F}_i) \geqslant \mathrm{OPT}_k^{(2,2)}(P_\rho) + (ts + t - 1)(R - k)$ by the bound on the change in cost from modifying the color points. Choosing as centers the vectors $\mathbf{e}_{N+c}$ for $c \in [k]$ in the original instance gives $\mathrm{OPT}_k^{(2,2)}(P_\rho) - (ts + t - 1)(m - R) \leqslant 0$, giving us

$$\begin{aligned}\mathrm{OPT}_k^{(2,1)}(Q_\rho) &\geqslant rm + \frac{1}{2r}\left(\mathrm{OPT}_k^{(2,2)}(P_\rho) - (ts + t - 1)(m - R)\right) \\ &\quad - O_t\left(\frac{s^2kd}{T^3}\right) - 2Tk - \frac{(ts + t - 1)k}{2r}.\end{aligned}$$

Since $m = |P_\rho|$, rearranging gives that

$$\begin{aligned}&\mathrm{OPT}_k^{(2,1)}(Q_\rho) - \left(r - \frac{ts + t - 1}{2r}\right)|P_\rho| - \frac{\mathrm{OPT}_k^{(2,2)}(P_\rho) + (ts + t - 1)R}{2r} \\ &\geqslant -O_t\left(\frac{s^2kd}{T^3}\right) - 2Tk - \frac{(ts + t - 1)k}{2r} \geqslant -\frac{skd}{2r}\eta.\end{aligned}$$

For the last inequality, we first choose $T/\sqrt{s}$ sufficiently large, depending only on t and η, so that $O_t(s^2kd/T^3)$ is at most $\eta skd/(4r)$. With this choice fixed, $r = \Theta(T)$ and $d = \omega(1)$ imply $2Tk + (ts + t - 1)k/(2r) = o(skd/r)$, completing the proof.

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